



Bahir Dar University
Bahir Dar Institute of Technology
Faculty of Computing

Requirement Analysis Document (RAD)
for
Industrial project on Medicine Price And Availability Finder Platform

Submitted to the faculty of computing in partial fulfillment of the requirements for the degree
of Bachelor of Science in information technology

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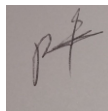
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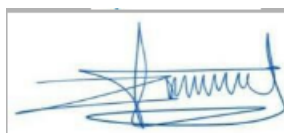
Declaration

The Project is our own and has not been presented for a degree in any other university and all the sources of material used for the project have been duly acknowledged.

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Faculty: Computing

Program: Information Technology

Project Title: **Medicine Price And Availability Finder Platform**

This is to certify that I have read this project and that in my supervision and the students' performance, it is fully adequate, in scope and quality, as a project for the degree of Bachelor of Science.

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It is approved that this project has been written in compliance with the formatting rules laid down by the faculty.

Roles and Responsibilities of the Group Members

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Database Design		√
UI/UX Design	√	√
Implementation	√	√
Testing and Debugging		√
Documentation	√	
Presentation Preparation	√	√
Final Review and Consensus Discussion	√	√

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This project is dedicated to all those who contributed directly or indirectly to its success.

List of Acronyms

Acronym	Full Meaning
RAD	Requirement Analysis Document
API	Application Programming Interface
OTP	One-Time Password
DBMS	Database Management System
UML	Unified Modeling Language
GPS	Global Positioning System
UI	User Interface
CDE	Continuing and Distance Education
CRUD	Create, Read, Update, Delete
FREQ	Functional Requirement
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
ERD	Entity Relationship Diagram
PK	Primary Key
FK	Foreign Key
UAT	User Acceptance Testing
IT	Information Technology
NoSQL	Not Only SQL
FCM	Firebase Cloud Messaging
WHO	World Health Organization

List of Figures

- Figure 2.1: Use Case Diagram – System Actors and Interactions
- Figure 2.2: User Interface Prototype – Medicine Search Screen
- Figure 2.3: State Chart Diagram – Prescription Lifecycle
- Figure 2.4: Activity Diagram – Medicine Search Workflow
- Figure 2.5: Sequence Diagram – Prescription Issuance
- Figure 2.6: Analysis Class Diagram – System Entities
- Figure 2.7: Conceptual Class Diagram – System Entities
- Figure 2.8: User Interface Prototype – Patient Dashboard
- Figure 3.1: System Architecture Diagram
- Figure 3.2: System component Diagram
- Figure 3.3: deployment diagram
- Figure 3.4: Design diagram
- Figure 3.5: Persistent diagram
- Figure 3.6: user-interface diagram

List of Tables

- Table 2.1: Functional Requirements
- Table 2.2: Business Rules
- Table 2.3: Logic Models
- Table 2.4: Non-Functional Requirements
- Table 2.5: Hardware Requirements
- Table 2.6: Software Requirements
- Table 2.7: CRC Analysis
- Table 2.8: Change Cases

Table of Contents (extracted from the document)

Section	Title	Page
Declaration		i
Roles and Responsibilities of the Group Members		ii
Acknowledgment		iii
List of acronyms		iv
List of Figures		v
List of Tables		vi
Table of Contents		vii
Abstract		x
Chapter One: Introduction		1
1.1	Background	2
1.2	Statement of the problem	3
1.3	Objectives	3
1.3.1	General Objective	3
1.3.2	Specific Objectives	3
1.4	Methodology	4
1.4.1	Requirement gathering methods	4
1.4.2	Analysis and design Methodology	4
1.4.3	Implementation Methodology	5
1.5	Feasibility	5
1.6	Beneficiaries or significant of the project	6
1.7	Limitations of the project	7
1.8	Scope of the project	8
1.9	Organization of the project	9
Chapter Two: System features.		10
2.1	The Existing System	10
2.2	Proposed System	11
2.3	Requirement Analysis	12
2.3.1	Functional requirement	12
2.3.2	System Use case	12
2.3.3	Business Rule Documentation	18
2.3.4	User Interface prototype	19
2.3.5	State chart diagram	20
2.3.6	Activity Diagram	21
2.3.7	Sequence diagram	22
2.3.8	Analysis Class Model	23
2.3.9	Logic model	24
2.4	Non functional requirement	25
2.5	System Requirement	27
2.5.1	Hardware requirements	27
2.5.2	Software requirements	28
2.6	Key abstraction with CRC analysis	29

	Section	Title	Page
2.6.1		Conceptual modeling: Class diagram	29
2.6.2		Identifying change cases	31
2.6.3		User Interface Prototyping	31
3		Chapter 3: System Design	34
3.1		Architectural Design	34
3.1.1		Component Modeling	35
3.1.2		Deployment Modeling	36
3.2		Detail Design	37
3.2.1		Design Class Model	37
3.2.2		Persistent Model	38
3.3		User Interface Design	39
3.4		Access Control and Security	40
--		References	43
--		Appendices	44

Abstract

The Medicine Price and Availability Finder Platform is a cross-platform digital solution designed to improve access to essential medicines, reduce medication errors, and increase transparency in Ethiopia's pharmaceutical supply chain. Built with Flutter for mobile and web and Firebase for real-time data services, the platform connects patients, pharmacies, healthcare professionals, and administrators in a single, role-based system. Key features include a searchable medicine catalog, real-time inventory and price updates by pharmacies, Google Maps integration for locating nearby outlets, clinician-issued verified digital prescriptions (token/QR), pharmacist dispensing workflows with standardized medication guidance, and user reporting and administrative moderation. The project responds to documented gaps—high rates of prescription and administration errors, fragmented price information, and time-consuming manual searches—by enabling verified clinicians to issue secure prescriptions, allowing pharmacists to record dispensing actions and guidance, and providing patients with timely notifications about availability and price changes. The development approach uses object-oriented analysis and agile implementation over a six-month timeline, relying on free-tier and open-source tools for pre-deployment development and estimating modest post-deployment cloud costs. The platform prioritizes security and privacy through Firebase Authentication, role-based access control, encrypted prescription storage, and immutable audit logs. Pilot deployment, training, and evaluation are planned in selected cities, with sustainability pursued through partnerships and service agreements with hospitals and pharmacies (patients access the service free). By centralizing inventory data and integrating prescription verification, the platform aims to reduce unnecessary travel and out-of-pocket costs, lower preventable medication harm, and support Ethiopia's broader digital health objectives.

Chapter One: Introduction

1.1 Background

Access to accurate, affordable, and up-to-date information about medicine prices and availability remains a major challenge in Ethiopia. In many cases, patients are forced to contact or physically visit multiple pharmacies simply to locate prescribed medicines, compare prices, or confirm whether a drug is in stock. As a result, the process becomes time-consuming, costly, and inefficient. This burden is even heavier for patients with chronic illnesses or those facing urgent medical needs, where delays can have serious consequences.

It is also worth noting that medication errors represent a significant public health concern in the country. Approximately 57.6% of patients experience medication errors during care, while about 55.8% of prescription sheets contain errors. These errors commonly include omission of medicines, incorrect dosages, or the use of wrong drug combinations. Beyond that, administration-related issues such as missed doses, incorrect timing, and poor documentation further increase the risk to patients.

Hospital-based studies provide additional insight into the severity of the problem. In pediatric wards, there are 9.2 adverse drug events per 100 admissions, with 35% of these events considered preventable. Similarly, emergency wards report medication errors in 74.4% of patients, with nearly 40% resulting in potentially moderate harm. On a broader scale, medication errors contribute to hundreds of thousands of preventable deaths globally each year and remain a leading cause of avoidable harm in East Africa.

Previous platforms such as HuluMed.org and FindKuta.com attempted to address challenges related to medicine access. However, these platforms were incomplete, non-functional, or lacked nationwide real-time coverage. More importantly, they did not support prescription verification, provide medication guidance, or contribute meaningfully to medication error reduction.

Essentially, this project proposes a Medicine Price and Availability Finder Platform designed to address these gaps. The system serves as a unified digital platform that provides real-time medicine price and stock information while supporting secure prescription management and professional-verified medical records. By integrating patients, pharmacies, and healthcare professionals, the platform aims to improve accessibility, reduce medication errors, and enhance patient safety across the nation.

1.2 Statement of the Problem

The absence of a centralized digital system for medicine price and availability in Ethiopia has created significant challenges for both patients and healthcare providers. In practice, patients are often forced to physically visit multiple pharmacies to locate prescribed medicines, compare prices, or confirm availability. This process leads to unnecessary delays in treatment and increases out-of-pocket costs, particularly for those who require timely access to medication.

It is also worth noting that medication errors are widespread within the healthcare system. Prescription sheets frequently contain omissions, incorrect dosages, or unsafe drug combinations. Beyond that, administration errors such as missed doses, incorrect timing, and poor documentation further compromise patient safety and undermine the quality of care.

Existing platforms have attempted to address some of these challenges, but they have failed to provide comprehensive, real-time, and nationwide solutions. In many cases, these systems lack prescription verification, medication guidance, and proper integration with healthcare professionals. As a result, critical gaps remain in transparency, efficiency, and patient safety.

Essentially, the proposed system seeks to address these issues by offering a centralized, real-time platform that integrates pharmacies, patients, and healthcare professionals. By doing so, it aims to improve medicine accessibility, reduce medication errors, and enhance overall healthcare outcomes.

1.3 Objectives

1.3.1 General Objective

To design and develop a centralized Medicine Price and Availability Finder Platform that provides real-time access to medicine stock and pricing information.

1.3.2 Specific Objectives

- To develop a mobile and web-based application that allows users to search for medicines by name and location.
- To develop a mobile and web-based application that allows users to search for medicines by name and location.
- To provide real-time information on medicine prices and availability across registered pharmacies.
- To design and implement secure OPT-based user authentication for patients, pharmacies, healthcare professionals, and administrators.
- To enable healthcare professionals to issue digital prescriptions through the system.
- To allow pharmacies to validate prescriptions and manage medicine stock and pricing within their authorized scope.
- To integrate location-based services that help users identify nearby pharmacies offering required medicines.
- To implement a notification mechanism that informs users about prescription updates and changes in medicine availability.
- To design a user-friendly interface that allows users to perform core system functions efficiently
- To test and evaluate the system in terms of functionality, usability, security, and performance.

1.4 Methodology

1.4.1 Requirement Gathering Methods

To ensure the system meets user needs and effectively addresses existing healthcare challenges, multiple requirement gathering techniques were employed. Interviews were conducted with patients, pharmacists, and healthcare professionals to better understand their pain points, daily experiences, and expectations from a digital solution. In addition, observation was carried out within pharmacy settings, focusing on workflows and patient interactions in order to identify inefficiencies, delays, and recurring issues in medicine access.

It is also worth noting that document analysis played a key role in the process. Prescription sheets, pharmacy records, and healthcare reports were reviewed to identify common errors, missing information, and systemic gaps. Beyond that, questionnaires were distributed to both patients and pharmacists to collect quantitative data related to medicine accessibility, availability, and pricing challenges. A literature review was also undertaken, analyzing prior studies and reports on medication errors and healthcare digitalization in Ethiopia to provide contextual and evidence-based support for the system requirements.

Requirement modeling was performed using an Object-Oriented (OO) approach. This approach was selected because it allows real-world entities to be represented clearly, supports modular system design, and facilitates scalability as system requirements evolve over time.

1.4.2 Analysis and Design Methodology

The project adopts Object-Oriented Analysis and Design (OOAD) methodology to ensure modularity, scalability, and maintainability throughout the system development process. In this approach, use case modeling is applied to define system functionalities and clearly outline interactions between actors and the system. Activity diagrams are used to visualize key workflows such as medicine search, prescription issuance, and stock updates, helping to clarify process flow and decision points.

Beyond that, sequence diagrams are employed to map interactions between system components and external actors over time, while class diagrams represent core system entities, their attributes, and the relationships between them. Data Flow Diagrams (DFD) are also utilized to illustrate how data moves across the system, ensuring a clear understanding of data inputs, processing, and outputs. In addition, prototyping is carried out to develop user interface prototypes, allowing usability validation and the collection of early user feedback.

Tools used for analysis and design include UML modeling tools such as Visual Paradigm and Edraw Max, Microsoft Word for documentation purposes, and Microsoft PowerPoint for presentations.

1.4.3 Implementation Methodology

The system will be implemented using modern mobile and web technologies, with a clear focus on rapid development, scalability, and long-term sustainability. To achieve cross-platform compatibility, Flutter will be used as the primary development framework, enabling the creation of mobile and web applications from a single codebase.

It is also worth noting that Firebase Firestore will serve as the real-time database, supporting efficient storage and synchronization of medicine stock data, prescriptions, and user information. For secure access, Firebase Authentication will be employed to provide reliable login functionality and enforce role-based access control for patients, pharmacists, and healthcare professionals. Beyond that, the system will integrate external services through APIs, including Google Maps API for pharmacy location and navigation services, as well as Firebase Cloud Messaging for delivering real-time notifications to users.

The project timeline spans a total of six months and is structured into clearly defined phases. Phase 1 focuses on requirement gathering and analysis and will be completed within one month. Phase 2 covers system design and prototyping, also scheduled for one month. Phase 3 involves core system development and will take three months, followed by Phase 4, which includes testing, deployment, and documentation over a one-month period.

In terms of budget, the project will be developed with a Birr 0 initial cost by relying on open-source tools and free-tier cloud services. Revenue generation is planned after deployment by charging hospitals and pharmacies for system usage, while patients will continue to access the platform free of charge.

1.5 Feasibility

Feasibility analysis is conducted to determine whether the proposed Medicine Price and Availability Finder Platform can be successfully developed and implemented within the available resources, time, and technical capacity. This analysis helps assess the practicality of the project before implementation. The feasibility study is categorized into three aspects: economic feasibility, technical feasibility, and time feasibility.

Economic Feasibility

The project is economically feasible because it requires no initial budget. Development will rely on open-source tools and free-tier cloud services such as Flutter SDK, Firebase Firestore, and Google Maps API. As a result, licensing costs are eliminated, and financial barriers to development are significantly reduced.

- Hospitals and pharmacies will be charged for system usage after deployment, ensuring long-term sustainability and steady revenue generation.
- Patients will access the platform free of charge, which increases adoption and improves accessibility to essential medicine information.
- The system will reduce costs associated with manual processes such as printing, physical record-keeping, and repeated patient visits to multiple pharmacies.

Technical Feasibility

The project is technically feasible because the required technologies and expertise are readily

available. The selected technologies are widely used, reliable, and suitable for the system's functional requirements.

- **Development Framework:** Flutter ensures cross-platform compatibility for both mobile and web applications.
 - **Database:** Firebase Firestore provides scalable, real-time data storage and synchronization.
 - **Authentication:** Firebase Authentication ensures secure login and role-based access control.
 - **APIs:** Google Maps API supports location-based services for pharmacy discovery.
 - The development team possesses the necessary skills in mobile and web development, database management, and API integration.
 - The system requires only standard hardware and internet connectivity, making it accessible to pharmacies and hospitals nationwide.

Time Feasibility

The project is time-feasible, with a clearly defined six-month development timeline that aligns with academic requirements.

- Month 1: Requirement gathering and analysis.
- Month 2: System design and prototyping.
- Months 3–5: Core development of mobile and web applications.
- Month 6: Testing, deployment, and documentation.

By focusing on core functionalities such as medicine search, stock updates, prescription management, and notifications, the project scope remains manageable within the academic timeframe. Beyond that, advanced features such as analytics and insurance integration are intentionally excluded from the first release to avoid delays and ensure timely completion.

1.6 Beneficiaries / Significance of the Project

The Medicine Price and Availability Finder Platform is designed to deliver meaningful benefits to multiple stakeholders within Ethiopia's healthcare ecosystem. Its significance extends beyond individual users, influencing patients, pharmacies, healthcare professionals, hospitals, and the broader public health sector in practical and measurable ways.

Patients

Patients will gain real-time access to information on medicine availability and pricing, allowing them to make informed decisions without unnecessary delays. As a result, they can save both time and money by avoiding repeated physical visits to multiple pharmacies. It is also worth noting that patients will receive timely notifications when medicines become available or when prices change. In addition, access to verified prescriptions and clear medication usage instructions will help reduce the risk of misuse and improve adherence to treatment.

Pharmacies

Pharmacies will benefit from improved transparency by providing accurate and up-to-date stock and price information. This visibility can attract more customers and strengthen their competitive position. Beyond that, digital prescription validation will help streamline workflows and reduce

manual errors. Being part of a verified and centralized system also allows pharmacies to build trust with patients while reducing administrative burdens through automated stock management and reporting.

Healthcare Professionals

Healthcare professionals will be able to issue secure and verified digital prescriptions, which plays a key role in reducing medication errors. The platform also supports the maintenance of accurate patient medical records within a centralized system. In practice, this enables professionals to provide standardized medication guidance, including dosage, frequency, duration, and timing, ultimately enhancing patient safety and improving treatment outcomes.

Hospitals

Hospitals will experience improved efficiency in medicine distribution and prescription handling. This contributes to a reduction in medication-related risks and strengthens patient safety measures. It is also worth noting that centralized data access supports better monitoring and reporting, while costs associated with manual record-keeping and inefficient workflows are reduced.

Government and Public Health Sector

For the government and public health sector, the platform supports Ethiopia's digital health transformation goals by enabling data-driven healthcare services. It improves nationwide access to essential medicines and enhances transparency within the medicine supply chain. Beyond that, reducing preventable medication errors contributes to better public health outcomes and provides a foundation for future integration with national health systems and policies.

Overall Significance

Overall, the project enhances patient safety by reducing medication errors and promotes transparency and trust in medicine pricing and availability. It also contributes to cost reduction for patients, pharmacies, and hospitals. Essentially, the platform aligns with national healthcare digitalization strategies, supporting a modern, efficient, and more equitable healthcare delivery system.

1.7 Limitations of the Project

Despite the significant benefits offered by the proposed Medicine Price and Availability Finder Platform, certain limitations exist due to technical, infrastructural, and contextual constraints. These limitations may affect system performance, accessibility, and adoption, particularly during the initial implementation phase.

- **Internet Dependency:** The system requires stable internet connectivity to support real-time updates and data synchronization. As a result, accessibility may be limited in rural or remote areas where internet coverage is poor or unavailable.
- **Data Accuracy:** The reliability of the platform depends on pharmacies and healthcare professionals providing accurate and timely data. Delayed updates or incorrect entries can reduce trust in the information presented to users.
- **Adoption Challenges:** Some pharmacies and hospitals may be slow to adopt the system due to resistance to change, limited technical skills, or existing infrastructure constraints.

- **Budget Constraints:** Due to limited resources, advanced features such as integration with insurance systems, predictive analytics, or offline access may not be included in the first release.
- **Security Risks:** Although secure authentication and encryption mechanisms are implemented, potential cyber threats remain and require continuous monitoring, regular updates, and ongoing security management.
- **Limited Scope in First Release:** The initial version focuses on core functionalities, including medicine search, stock updates, prescription management, and notifications. Features such as online purchasing, advanced analytics, and integration with national health systems are intentionally excluded to ensure timely and stable delivery.

- **1.8 Scope of the Project**

The scope of the Medicine Price and Availability Finder Platform defines the boundaries of what the system will and will not include in its first release. It is intended to keep the project focused, achievable within the given timeframe, and aligned with the core objectives of improving medicine accessibility and reducing prescription errors.

Included in the Scope

- **Medicine Search:** Patients can search for medicines by name, brand, or category, making it easier to locate required medications.
- **Real-Time Stock and Price Updates:** Pharmacies update medicine availability and pricing instantly, ensuring users have access to current and reliable information.
- **Prescription Management:** Healthcare professionals issue secure and verified digital prescriptions through the system.
- **Notifications:** Patients receive alerts when medicines become available or when prices change, supporting timely decision-making.
- **Location Services:** Integration with Google Maps API helps patients locate nearby pharmacies and obtain directions.
- **Multilingual Interface:** The platform supports multiple languages to improve accessibility for users across Ethiopia.
- **Role-Based Access Control:** Administrators verify pharmacies and healthcare professionals, ensuring trust, accountability, and transparency within the system.
- **Medication Guidance:** Pharmacists provide standardized instructions on dosage, frequency, duration, and timing to promote safe medicine use.

Excluded from the Scope

- **Direct Online Medicine Purchases:** The system will not support e-commerce transactions for medicines in its initial release.

- **Insurance System Integration:** Linking the platform with health insurance providers is excluded to reduce system complexity.
- **Advanced Analytics:** Predictive analytics and demand forecasting features are not included in the first version.
- **Offline Access:** The system requires active internet connectivity and does not support offline functionality during the initial release.

1.9 Organization of the Project

This project is organized into chapters and sections to present the study in a clear, logical, and academically rigorous manner. Each chapter focuses on a specific aspect of the **Medicine Price and Availability Finder Platform** and builds upon the previous sections to provide a comprehensive understanding of the system.

Chapter One: Introduction

- Provides the background of the study and outlines the challenges in medicine accessibility and prescription management.
- States the problem, objectives, methodology, feasibility, beneficiaries, limitations, scope, and organization of the project.
- Establishes the foundation for the need and relevance of the proposed system.

Chapter Two: System Features

- Describes the existing system and its drawbacks.
- Presents the proposed system, its functionalities, and advantages.
- Includes requirement analysis, functional and non-functional requirements, use cases, business rules, user interface prototypes, and UML diagrams (state chart, activity, sequence, class, logic models).
- Defines system requirements (hardware and software) and key abstractions with CRC analysis.

Chapter Three: System Design

- Provides detailed design models of the system using UML diagrams.
- Explains the architecture, data flow, and interaction between system components.
- Includes conceptual modeling, change cases, and user interface design considerations.

Chapter Four: Implementation

- Discusses the tools, frameworks, and technologies used for system development.
- Explains database design, API integration, authentication, and notification services.
- Outlines the development phases and testing strategies.

Chapter Five: Results and Discussion

- Presents the outcomes of the system implementation.
- Evaluates system performance, usability, and reliability.

- Discusses user feedback and compares the proposed system with the existing manual process.

Chapter Six: Conclusion and Recommendations

- Summarizes the key findings of the project.
- Highlights the contributions of the system to healthcare digitalization.
- Provides recommendations for future improvements and expansions (e.g., insurance integration, analytics, offline access).

References

- Lists all sources consulted during the project, including books, research papers, reports, and online resources.

Appendices

- Contains supporting documents such as questionnaires, interview guides, sample forms, prototype screenshots, and sample datasets.

Chapter Two: System Features

2.1 The Existing System

The current healthcare environment in Ethiopia lacks a centralized and automated platform for medicine price and availability. Patients, pharmacies, and healthcare professionals rely on fragmented and manual processes that are inefficient, error-prone, and costly.

2.1.1 Description of the Existing System

- Patients must physically visit multiple pharmacies or make phone calls to locate prescribed medicines.
- Medicine prices vary widely between pharmacies, with no transparent mechanism for comparison.
- Prescription sheets are handwritten and often contain errors such as omissions, incorrect dosages, or unsafe drug combinations.
- Pharmacies maintain stock records manually, leading to inconsistencies and delays in updating availability.
- Healthcare professionals issue prescriptions without centralized verification, increasing the risk of medication errors.
- Patients receive little or no guidance on proper medication usage, dosage, and timing.

2.1.2 Step-by-Step Operation of the Existing System

1. Patient receives a handwritten prescription from a healthcare professional.
2. Patient visits one or more pharmacies to check medicine availability and price.
3. Pharmacy staff manually check stock records and provide information.
4. If medicine is unavailable, the patient repeats the process at another pharmacy.
5. Patient purchases medicine without standardized guidance on usage.
6. Prescription and purchase records remain paper-based and fragmented.

2.1.3 Drawbacks of the Existing System

- **Time-Consuming:** Patients spend significant time visiting multiple pharmacies.
- **Costly:** Lack of transparency forces patients to pay higher prices.
- **Error-Prone:** Handwritten prescriptions and manual records increase the risk of medication errors.
- **Data Inconsistency:** Stock records are unreliable and not updated in real time.
- **Limited Accessibility:** Patients in rural areas face greater challenges due to poor communication and infrastructure.
- **No Centralized Verification:** Prescriptions are not validated against standardized medical guidelines.
- **Poor Patient Guidance:** Patients often misuse medicines due to lack of proper instructions.

2.2 Proposed System

The proposed **Medicine Price and Availability Finder Platform** is a centralized, web and mobile- based solution designed to overcome the inefficiencies and risks of the existing manual system. It integrates patients, pharmacies, healthcare professionals, and administrators into a unified digital ecosystem that ensures transparency, accessibility, and safety in medicine distribution and usage.

2.2.1 Description of the Proposed System

The system focuses on automating and modernizing medicine accessibility and prescription management. It introduces:

- **Real-Time Medicine Tracking:** Pharmacies update stock and prices instantly, ensuring patients access accurate information.
- **Secure Prescription Management:** Healthcare professionals issue verified digital prescriptions, reducing medication errors.
- **Location Services:** Integration with Google Maps API allows patients to locate nearby pharmacies and obtain directions.
- **Notifications:** Patients receive alerts when medicines become available or when prices change.
- **Multilingual Interface:** The platform supports multiple languages to ensure accessibility across Ethiopia.
- **Role-Based Access Control:** Administrators verify pharmacies and healthcare professionals, ensuring trust and transparency.
- **Medication Guidance:** Pharmacists provide standardized instructions on dosage, frequency, duration, and timing.

2.2.2 How the Proposed System Works

1. **Patient Registration:** Patients create accounts securely using Firebase Authentication.
2. **Medicine Search:** Patients search medicines by name, brand, or category.
3. **Stock and Price Updates:** Pharmacies update medicine availability and pricing in real time.
4. **Prescription Issuance:** Healthcare professionals issue secure, verified prescriptions stored in the system.

5. **Notifications:** Patients receive alerts about medicine availability and price changes.
6. **Location Services:** Patients use Google Maps integration to find nearby pharmacies.
7. **Medication Guidance:** Pharmacists provide standardized instructions to ensure safe medicine usage.
8. **Administration:** System administrators verify pharmacies and healthcare professionals to maintain trust and transparency.

2.2.3 Advantages of the Proposed System

- **Efficiency:** Reduces manual work and processing time for patients and pharmacies.
- **Accuracy:** Improves data reliability and reduces medication errors.
- **Accessibility:** Provides real-time information and eliminates unnecessary physical travel.
- **Transparency:** Ensures patients can compare medicine prices across pharmacies.
- **Cost Reduction:** Minimizes expenses related to manual processes and repeated visits.
- **Scalability:** Supports future expansion, including integration with insurance systems and advanced analytics.
- **Patient Safety:** Enhances prescription verification and medication guidance, reducing risks of misuse.

2.3 Requirement Analysis

Requirement analysis defines the functional and non-functional needs of the **Medicine Price and Availability Finder Platform**. It ensures the system meets user expectations, addresses existing challenges, and provides a foundation for design and implementation.

2.3.1 Functional Requirements

Functional requirements specify the services and operations the system must perform.

- **Patient Registration & Authentication:** Patients register securely using Firebase OTP.
- **Medicine Search:** Patients search medicines by name, brand, or category.
- **Stock & Price Updates:** Pharmacies update medicine availability and pricing in real time.
- **Prescription Management:** Healthcare professionals issue secure, verified prescriptions.
- **Notifications:** Patients receive alerts about medicine availability and price changes.
- **Location Services:** Integration with Google Maps API to locate nearby pharmacies.
- **Role-Based Access Control:** Patients, pharmacies, healthcare professionals, and administrators have distinct privileges.
- **Medication Guidance:** Pharmacists provide standardized instructions (dosage, frequency, duration, timing).

ID	Requirement	Description	Actors Involved
FR-01	Patient Registration & Authentication	Patients must be able to register and log in securely using Firebase OTP verification.	Patient, System
FR-02	Medicine Search	Patients can search medicines by name, brand, or category.	Patient, System
FR-03	Stock & Price	Pharmacies must update medicine	Pharmacy, System

ID	Requirement	Description	Actors Involved
	Updates	availability and pricing in real time.	
FR-04	Prescription Management	Healthcare professionals issue secure, verified digital prescriptions linked to patient accounts.	Healthcare Professional, Patient, System
FR-05	Notifications	Patients receive alerts when medicines become available or when prices change.	Patient, System
FR-06	Location Services	Integration with Google Maps API to locate nearby pharmacies and provide directions.	Patient, System
	Role-Based Access	Different roles (Patient, Pharmacy,	
FR-07	Control	Healthcare Professional, Administrator) must have distinct privileges. Pharmacists provide standardized	All Actors
FR-08	Medication Guidance	instructions (dosage, frequency, duration, timing) for safe medicine usage.	Pharmacy, Patient
	Verification of	Administrators verify pharmacies and healthcare professionals before granting access.	Administrator,
FR-09	Pharmacies & Professionals	System must store and manage medicine, prescription, and user data securely in Firebase Firestore.	Pharmacy, Healthcare Professional
FR-10	Data Management		System

Table 2.1: Functional Requirements

2.3.2 System Use Case

2.3.2.1 Use Case Diagram

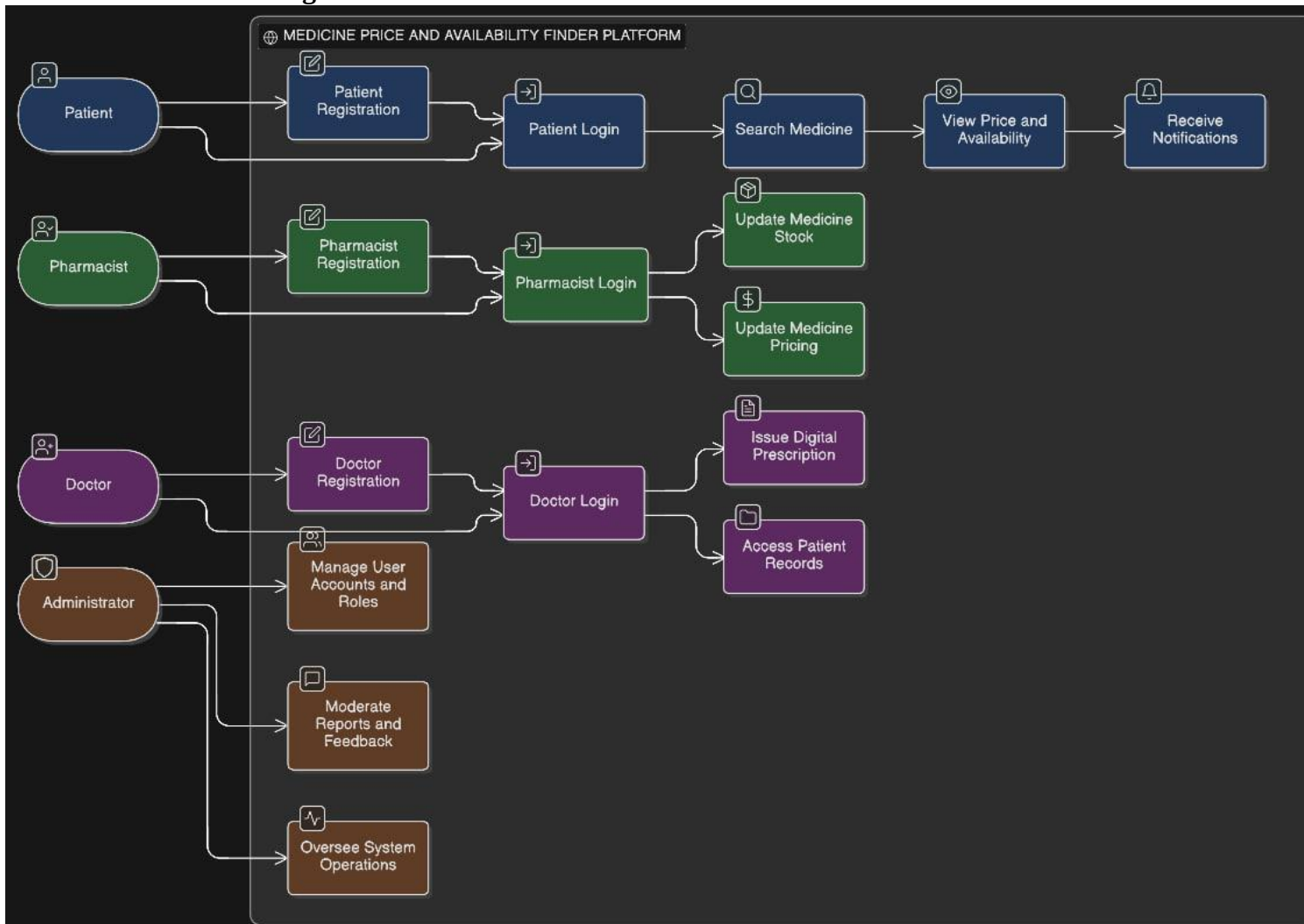


Figure 2.1: Use Case Diagram – System Actors and Interactions

2.3.2.2 Use Case Documentation

Each use case is documented with actors, preconditions, triggers, workflows, and postconditions.

- **Manage Patient Account:** Patients register, update profiles, and authenticate securely.
- **Search Medicine:** Patients query medicines by name, brand, or category.
- **Update Medicine Stock:** Pharmacies update stock and pricing information.
- **Issue Prescription:** Healthcare professionals issue verified prescriptions linked to patient accounts.
- **Notify Patient:** System sends alerts about medicine availability and price changes.
- **Verify Pharmacy:** Administrators validate pharmacies and healthcare professionals before granting access.

Use Case ID	Use Case Name	Description	Primary Actor(s)	Secondary Actor(s)
UC-01	Manage Patient Account	Patients register, log in, and manage their profile securely.	Patient	System
UC-02	Search Medicine	Patients search medicines by name, brand, or category.	Patient	Pharmacy, System
UC-03	Update Medicine Stock	Pharmacies update medicine availability and pricing in real time.	Pharmacy	System
UC-04	Issue Prescription	Healthcare professionals issue	Healthcare	Patient, System

Use Case ID	Use Case Name	Description	Primary Actor(s)	Secondary Actor(s)
		secure, verified prescriptions linked to patients. System sends alerts about	Professional	
UC-05	Notify Patient	medicine availability and price changes.	System	Patient
UC-06	Verify Pharmacy/Professional	Administrators validate pharmacies and healthcare professionals before access.	Administrator	Pharmacy, Healthcare Professional
UC-07	Provide Medication Guidance	Pharmacists provide standardized dosage, frequency, and usage instructions.	Pharmacy	Patient
UC-08	Location Services	Patients locate nearby pharmacies using Google Maps integration.	Patient	System, Pharmacy

Table 2.2: Use Case Identification

Use Case ID	UC-01
Use Case Name	Manage Patient Account
Primary Actor	Patient
Secondary Actor(s)	System
Preconditions	Patient must have access to the application and internet connection.
Trigger	Patient initiates registration, login, or profile update.
Description	Patients register, authenticate, and manage their account securely using Firebase OTP verification.
Normal Flow	

Table 2.3: Use Case Description for Manage Patient Account

Patient opens the application.

1. Patient selects “Register” or “Login.”
2. System requests phone number/email for OTP verification.
3. Patient enters OTP received via SMS/email.
4. System verifies OTP and grants access.
5. Patient updates profile information (name, contact, medical history).
 - Invalid OTP entered → System prompts retry.
 - Network failure → System displays error message and requests reconnection.

Postconditions Patient account is created/updated and securely stored in the system. **Exceptions** OTP not delivered, incorrect credentials, or database connection failure.

Use Case ID	UC-02
Use Case Name	Search Medicine
Primary Actor	Patient
Secondary Actor(s)	Pharmacy, System
Preconditions	Patient must be logged into the system with a valid account.
Trigger	Patient initiates a search query for a medicine by name, brand, or category.
Description	Patients search for medicines in the system to view availability, pricing, and pharmacy location.

Table 2.4: Use Case Description for Search Medicine

Normal Flow

1. Patient opens the search interface.
 2. Patient enters medicine name, brand, or category.
 3. System queries the database for matching medicines.
 4. System retrieves stock and price information from pharmacies.
 5. System displays results with medicine details, availability, and pharmacy location.
 6. Patient selects a pharmacy to view directions and additional details.
 - No matching medicine found → System displays “Medicine not available.”
 - Medicine found but out of stock → System displays “Currently unavailable.”
 - Patient enters incomplete/incorrect query → System suggests possible matches.
- Postconditions** Patient receives accurate information about medicine availability, price, and pharmacy location.
- Exceptions** Database connection failure, network issues, or incorrect query input.

Use Case ID	UC-03
Use Case Name	Update Medicine Stock
Primary Actor	Pharmacy
Secondary Actor(s)	System
Preconditions	Pharmacy must be verified and logged into the system.
Trigger	Pharmacy staff initiates a stock or price update.
Description	Pharmacies update medicine availability and pricing in real time to ensure accurate patient information.

Table 2.5: Use Case Description for Update Medicine Stock

Normal Flow

1. Pharmacy staff logs into the system.
2. Staff navigates to the stock management dashboard.
3. Staff selects the medicine to update or adds a new medicine entry.
4. Staff enters updated details (price, quantity, availability).
5. System validates the input data.

6. System saves the updated information in the database.
 7. Patients are notified of changes in availability or price.
- Invalid data entered → System prompts correction.
 - Medicine not found in database → System requests new entry creation.
 - Network failure → System displays error and requests retry.

Postconditions Updated medicine stock and pricing are stored in the system and visible to patients.

Exceptions Database connection failure, incorrect data entry, or unauthorized pharmacy access.

Use Case ID	UC-04
Use Case Name	Issue Prescription
Primary Actor	Healthcare Professional
Secondary Actor(s)	Patient, System
Preconditions	Healthcare professional must be verified and logged into the system. Patient must have an active account.
Trigger	Healthcare professional initiates the process of issuing a prescription.
Description	Healthcare professionals issue secure, verified prescriptions linked to patient accounts, ensuring accuracy and accessibility.

Table 2.6: Use Case Description for Issue Prescription

Normal Flow

1. Healthcare professional logs into the system.
 2. Professional navigates to the prescription management dashboard.
 3. Professional selects the patient from the database.
 4. Professional enters medicine details (name, dosage, frequency, duration).
 5. System validates medicine information against the database.
 6. System generates a digital prescription linked to the patient's account.
 7. Patient receives notification of the new prescription.
- Invalid medicine entry → System prompts correction.
 - Patient not found → System requests verification or new patient registration.
 - Network failure → System displays error and requests retry.

Postconditions Prescription is securely stored in the system and accessible to both patient and pharmacy.

Exceptions Database connection failure, incorrect medicine details, or unauthorized healthcare professional access.

Use Case ID	UC-05
Use Case Name	Notify Patient

Use Case ID	UC-05
Primary Actor	System
Secondary Actor(s)	Patient
Preconditions	Patient must be registered and logged into the system.
Trigger	System detects a change in medicine availability, price, or prescription update.
Description	The system sends notifications to patients about medicine availability, price changes, or prescription updates.

Table 2.7: Use Case Description for Notify Patient

Normal Flow

1. System monitors medicine stock and price updates from pharmacies.
 2. System detects relevant changes (availability, price, prescription update).
 3. System generates a notification message.
 4. Notification is sent to the patient via the application or SMS/email.
 5. Patient views the notification in the dashboard or receives it on their device.
- Patient ignores notification → System stores it in notification history.
 - Patient acknowledges notification → System marks it as read.

Postconditions Patient is informed about medicine availability, price changes, or prescription updates.

Exceptions Network failure, notification delivery error, or incorrect patient contact information.

Use Case ID	UC-06
Use Case Name	Verify Pharmacy
Primary Actor	Administrator
Secondary Actor(s)	Pharmacy, System
Preconditions	Pharmacy must submit registration details to the system. Administrator must be logged in with verification privileges.
Trigger	Administrator initiates the process of verifying a pharmacy.
Description	Administrators validate pharmacy registration details to ensure authenticity before granting access to the system.

Table 2.8: Use Case Description for Verify Pharmacy

Normal Flow

1. Pharmacy submits registration details (name, license number, location, contact).
2. System stores the submitted details in the pending verification list.
3. Administrator logs into the system and navigates to the verification dashboard.
4. Administrator reviews pharmacy details and supporting documents.

5. Administrator approves or rejects the pharmacy registration.
 6. System updates the pharmacy status accordingly.
 7. Pharmacy receives notification of approval or rejection.
- Missing or invalid details → System prompts pharmacy to resubmit.
 - Suspicious or fraudulent details → Administrator rejects registration and flags pharmacy.
 - Network failure → System displays error and requests retry.

Postconditions Pharmacy is either verified and granted access or rejected with reasons provided.

Exceptions Database connection failure, incomplete registration details, or unauthorized administrator access.

2.3.3 Business Rule Documentation

Business rules define the policies and constraints under which the system operates.

- Only verified pharmacies can update medicine stock and prices.
- Prescriptions must be issued by licensed healthcare professionals.
- Patients can only access prescriptions linked to their account.
- Notifications are triggered when medicine availability or price changes occur.
- Administrators are responsible for verifying pharmacies and healthcare professionals.

Rule ID	Business Rule	Description
BR-01	Secure Authentication	All users (patients, pharmacies, healthcare professionals, administrators) must authenticate using Firebase OTP verification before accessing the system.
BR-02	Role-Based Access	Each actor has specific privileges: patients can search and view medicines, pharmacies can update stock, healthcare professionals can issue prescriptions, administrators can verify accounts.
BR-03	Verified Entities Only	Only verified pharmacies and healthcare professionals can interact with the system to ensure trust and authenticity.
BR-04	Real-Time Updates	Medicine stock and pricing must be updated in real time to maintain accuracy for patients.
BR-05	Prescription Integrity	Prescriptions must be digitally signed and linked to verified healthcare professionals to prevent fraud.
BR-06	Notification Delivery	Patients must be notified promptly about medicine availability, price changes, or prescription updates.
BR-07	Data Security	All patient, pharmacy, and prescription data must be stored securely in Firebase Firestore with encryption.
BR-08	Standardized Guidance	Pharmacists must provide standardized dosage, frequency, and usage instructions for medicines to ensure patient safety.
BR-09	Location Accuracy	Pharmacy locations must be integrated with Google Maps API to provide accurate directions.
BR-10	Audit & Compliance	All system activities (logins, updates, prescriptions, verifications) must be logged for auditing and compliance purposes.

Table 2.9: Business Rules

2.3.4 User Interface Prototype

Prototypes illustrate the design and layout of the system's user interface.

- **Patient Dashboard:** Search bar, medicine list, notifications panel, pharmacy map.
- **Pharmacy Dashboard:** Stock update form, price management, prescription validation.
- **Healthcare Professional Dashboard:** Prescription issuance, patient record management.
- **Administrator Dashboard:** Verification tools, system monitoring.

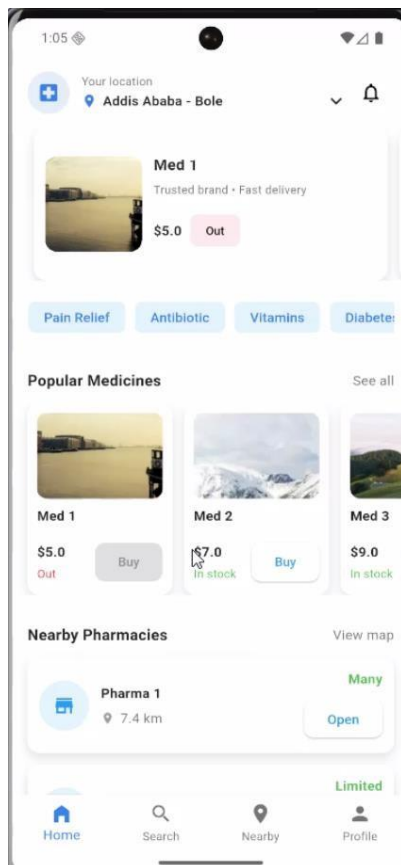


Figure 2.2: User Interface Prototype – Medicine Search Screen

2.3.5 State Chart Diagram

State chart diagrams represent the lifecycle of system entities.

- **Prescription Lifecycle:** Created → Verified → Issued → Used → Archived.
- **Medicine Lifecycle:** Added → Updated → Available → Out of Stock → Removed.

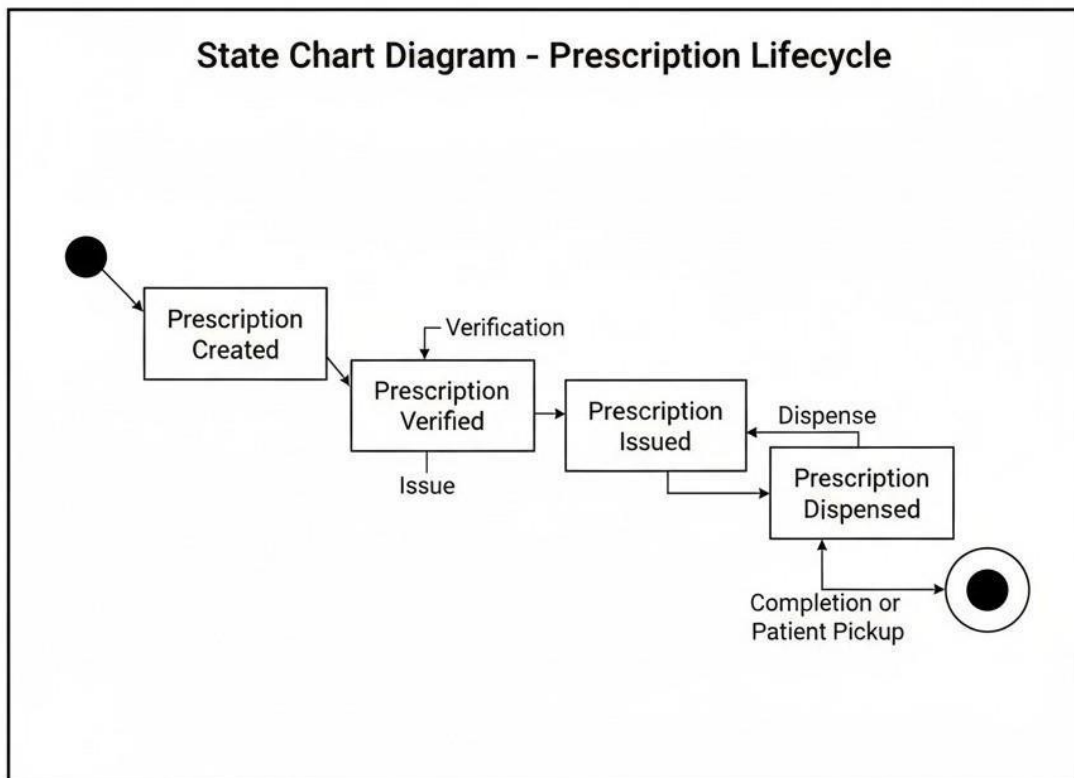


Figure 2.3: State Chart Diagram – Prescription Lifecycle

2.3.6 Activity Diagram

Activity diagrams illustrate workflows and processes.

- **Medicine Search Workflow:** Patient queries → System retrieves → Pharmacy updates → Results displayed.
- **Prescription Issuance Workflow:** Healthcare professional issues → System verifies → Patient notified.
- **Stock Update Workflow:** Pharmacy updates → System validates → Patients notified.

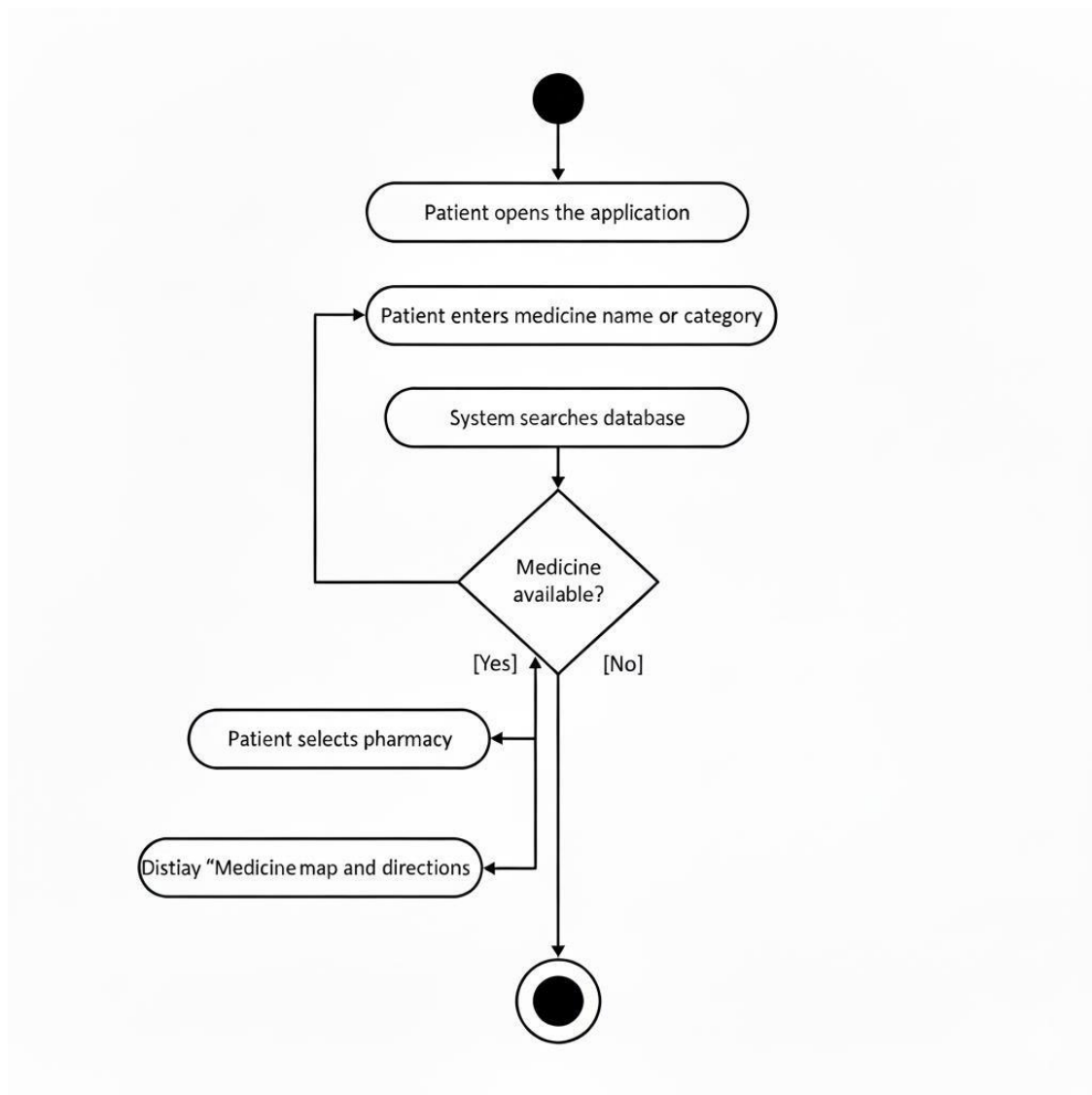


Figure 2.4: Activity Diagram – Medicine Search Workflow

2.3.7 Sequence Diagram

Sequence diagrams show interactions between system components over time.

- **Patient Searching for Medicine:** Patient → System → Pharmacy → Results.
- **Healthcare Professional Issuing Prescription:** Professional → System → Patient → Pharmacy.
- **Pharmacy Updating Stock:** Pharmacy → System → Patient (notification).

Prescription Issuance Process

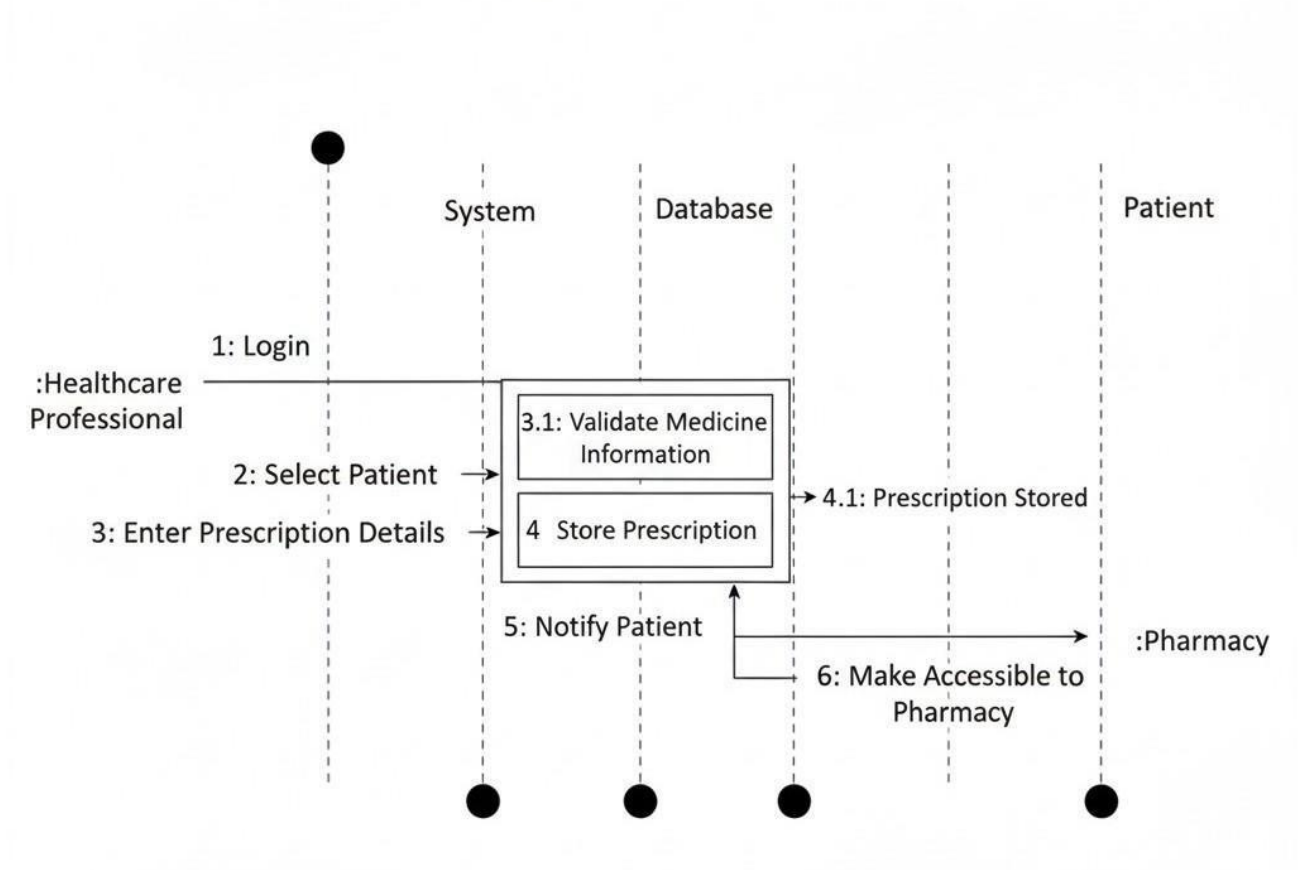


Figure 2.5: Sequence Diagram – Prescription Issuance

2.3.8 Analysis Class Model

Analysis class models represent system entities, attributes, and relationships.

- **Classes:** Patient, Pharmacy, Healthcare Professional, Administrator, Medicine, Prescription, Notification.
- **Relationships:**
 - Patient–Prescription (1..*)
 - Pharmacy–Medicine (1..*)
 - Healthcare Professional–Prescription (1..*)
 - Administrator–Pharmacy (verification)

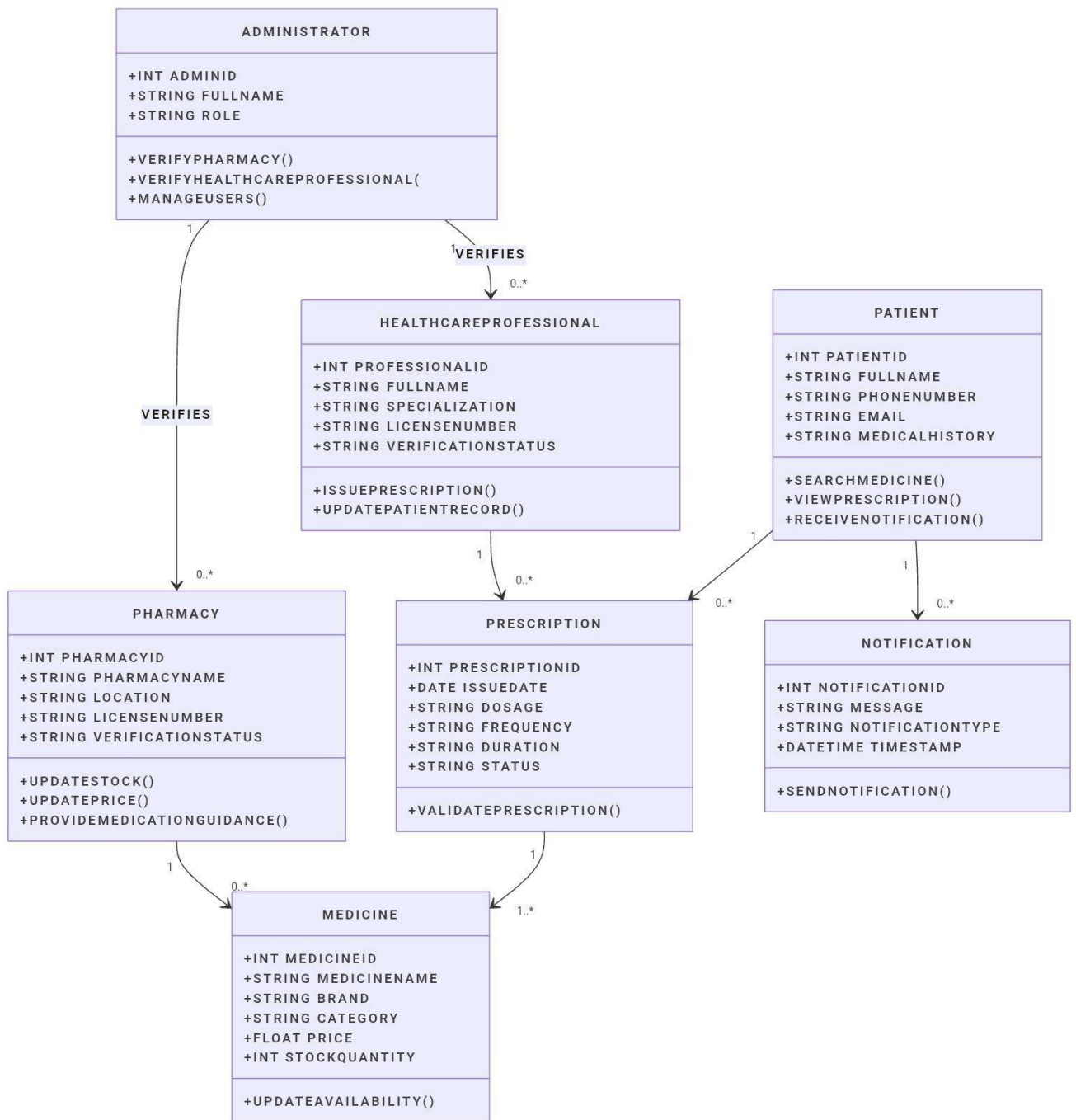


Figure 2.6: Analysis Class Diagram – System Entities

2.3.9 Logic Model

Logic models define the flow of data and decision-making rules.

- **Medicine Search Logic:** Patient query → Database lookup → Display results.
- **Prescription Validation Logic:** Professional issues → System verifies → Pharmacy validates.
- **Notification Trigger Logic:** Stock/price change → System detects → Patient notified.

Logic Model ID	Logic Model	Description
LM-01	Patient Registration Logic	Ensures secure patient registration and authentication using Firebase OTP verification. Validates user identity before granting access.
LM-02	Medicine Search Logic	Handles queries from patients, retrieves medicine data from the database, and displays availability, price, and pharmacy location.
LM-03	Stock Update Logic	Allows pharmacies to update medicine stock and pricing in real time, ensuring patients see accurate information.
LM-04	Prescription Issuance Logic	Validates healthcare professional credentials, links prescriptions to patient accounts, and stores them securely.
LM-05	Notification Logic	Monitors changes in medicine stock, price, or prescriptions and sends timely alerts to patients.
LM-06	Verification Logic	Administrators validate pharmacy and healthcare professional details before granting system access.
LM-07	Medication Guidance Logic	Ensures pharmacists provide standardized dosage, frequency, and usage instructions to patients.
LM-08	Location Services Logic	Integrates Google Maps API to provide accurate pharmacy locations and directions to patients.
LM-09	Role-Based Access Logic	Enforces access control rules, ensuring each actor (patient, pharmacy, healthcare professional, administrator) has appropriate privileges.
LM-10	Data Security & Compliance Logic	Encrypts sensitive data, logs system activities, and ensures compliance with auditing standards.

Table 2.10: Logic Models

2.4 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance standards, and operational constraints of the **Medicine Price and Availability Finder Platform**. These requirements ensure that the system is reliable, secure, and user-friendly, beyond its functional capabilities.

2.4.1 Performance Requirements

- The system must handle **at least 10,000 concurrent users** without performance degradation.
- Search results for medicine availability and pricing must be displayed within **3 seconds**.
- Notifications must be delivered to patients within **5 seconds** of stock or price updates.
- The system must support real-time synchronization between pharmacies, patients, and healthcare professionals.

2.4.2 Security Requirements

- All user data must be encrypted both in transit (SSL/TLS) and at rest.
- Authentication must be handled using **Firebase Authentication** with OTP verification.

- Role-based access control must ensure that only authorized users (patients, pharmacies, healthcare professionals, administrators) can access specific functionalities.
- Prescriptions must be digitally signed and verified to prevent tampering.
- The system must comply with healthcare data privacy standards.

2.4.3 Usability Requirements

- The interface must be intuitive and accessible to users with basic digital literacy.
- The system must support **multilingual interfaces** (e.g., Amharic, English, Afaan Oromo, Tigrigna).
- Navigation must be simple, with clear menus and search functionality.
- The mobile application must be compatible with both **Android and iOS** devices.
- Accessibility features must be included (e.g., font resizing, high-contrast mode).

2.4.4 Reliability Requirements

- The system must achieve **99.9% uptime** to ensure continuous availability.
- Automatic backup of data must occur daily to prevent data loss.
- The system must recover from failures within **2 minutes** using cloud-based redundancy.
- Notifications and updates must be consistent and reliable across all devices.

2.4.5 Scalability Requirements

- The system must scale horizontally to support additional pharmacies and hospitals nationwide.
- Database architecture must allow for expansion to millions of records without performance loss.
- APIs must be designed to integrate future services (e.g., insurance systems, analytics).

2.4.6 Maintainability Requirements

- The system must be modular, allowing updates and bug fixes without disrupting core services.
- Documentation must be provided for developers and administrators.
- Logs must be maintained for system activities to support troubleshooting.
- Updates must be deployed with minimal downtime using cloud CI/CD pipelines.

2.4.7 Portability Requirements

- The system must run on multiple platforms (web, Android, iOS).
- The application must be deployable on different cloud environments (Firebase, AWS, Azure).
- Data must be exportable in standard formats (CSV, JSON) for interoperability.

ID	Requirement	Description
NFR-01	Performance	The system must handle concurrent requests efficiently, ensuring search results are delivered within 3 seconds.

ID	Requirement	Description
NFR-02	Scalability	The system must support increasing numbers of users, pharmacies, and medicines without performance degradation.
NFR-03	Security	All data must be encrypted in transit and at rest, with secure authentication via Firebase OTP.
NFR-04	Availability	The system must maintain 99.9% uptime to ensure continuous access for patients and pharmacies.
NFR-05	Reliability	Medicine stock, price, and prescription data must remain consistent and accurate across all modules.
NFR-06	Usability	The interface must be intuitive, user-friendly, and accessible to patients, pharmacies, and healthcare professionals.
NFR-07	Maintainability	The system must allow easy updates, bug fixes, and enhancements without disrupting service.
NFR-08	Portability	The application must run smoothly across web and mobile platforms (Android/iOS).
NFR-09	Compliance	The system must adhere to healthcare data protection regulations and auditing standards.
NFR-10	Localization	The system must support multiple languages and regional settings for broader accessibility.

Table 2.11: Non-Functional Requirements

2.5 System Requirements

System requirements define the minimum and recommended hardware and software specifications necessary for the successful deployment and operation of the **Medicine Price and Availability Finder Platform**. These requirements ensure that the system runs efficiently, securely, and reliably across different environments.

2.5.1 Hardware Requirements

For End Users (Patients, Pharmacists, Healthcare Professionals):

- **Device:** Smartphone (Android/iOS) or personal computer with internet access.
- **Processor:** Minimum dual-core processor (1.8 GHz or higher).
- **RAM:** At least 2 GB (recommended 4 GB for smoother performance).
- **Storage:** Minimum 500 MB free storage for application installation and caching.
- **Display:** 5-inch screen or larger for mobile devices; 1024x768 resolution for desktops.
- **Internet Connectivity:** Stable 3G/4G/5G or broadband connection.

For Pharmacy/Hospital Administration Systems:

- **Processor:** Quad-core processor (2.5 GHz or higher).
- **RAM:** Minimum 8 GB (recommended 16 GB for handling large datasets).
- **Storage:** 250 GB HDD/SSD for local backups and logs.
- **Network:** Reliable broadband connection with at least 10 Mbps speed.
- **Peripheral Devices:** Barcode scanner (optional) for medicine stock management.

For Server/Cloud Hosting:

- **Processor:** Multi-core processor (8 cores or higher).
- **RAM:** Minimum 16 GB (recommended 32 GB for scalability).
- **Storage:** 1 TB SSD for database and file storage.
- **Network:** High-speed internet with redundancy and load balancing.
- **Backup:** Automated cloud backup and disaster recovery support.

ID	Hardware Requirement	Description
HR-01	Server Machine	High-performance server with minimum 16 GB RAM, quad-core processor, and 1 TB storage to host backend services and database.
HR-02	Client Devices	Smartphones (Android/iOS) and web browsers on PCs/laptops to access the platform.
HR-03	Network Equipment	Reliable routers, switches, and firewalls to ensure secure and stable connectivity between clients and servers.
HR-04	Backup Storage	External or cloud-based storage with at least 2 TB capacity for system backups and recovery.
HR-05	Power Supply	Uninterruptible Power Supply (UPS) to maintain system availability during power outages.
HR-06	Testing Devices	Multiple device types (low-end and high-end smartphones, tablets, PCs) for compatibility and performance testing.
HR-07	Security Hardware	Hardware firewalls and secure authentication devices to protect sensitive healthcare data.
HR-08	Internet Connectivity	High-speed broadband connection with redundancy to ensure continuous system access.

Table 2.12: Hardware Requirements

2.5.2 Software Requirements

For End Users (Patients, Pharmacists, Healthcare Professionals):

- **Operating System:** Android 8.0+ / iOS 12+ / Windows 10+ / macOS 10.14+.
- **Web Browser:** Latest versions of Chrome, Firefox, Safari, or Edge.
- **Mobile Application:** Flutter-based cross-platform app.
- **Additional Software:** PDF reader for prescription and report downloads.

For Pharmacy/Hospital Administration Systems:

- **Operating System:** Windows 10+ / Linux (Ubuntu 20.04+).
- **Database Client:** Firebase Firestore access tools.
- **Web Browser:** Chrome or Firefox (latest versions).
- **Utility Software:** Antivirus and firewall protection.

For Server/Cloud Hosting:

- **Operating System:** Linux (Ubuntu 20.04 LTS or higher).
- **Database:** Firebase Firestore (NoSQL) for real-time data storage.
- **Authentication:** Firebase Authentication for secure login and OTP verification.
- **Storage:** Firebase Storage and Cloudinary for file and image management.
- **APIs:** Google Maps API for location services; Firebase Cloud Messaging for notifications.
- **Development Tools:** Visual Studio Code, GitHub for version control, CI/CD pipelines for deployment.

ID	Software Requirement	Description
SR-01	Operating System	Server must run on Linux/Windows Server; client devices on Android, iOS, or modern web browsers.
SR-02	Database	Firebase Firestore for secure, scalable, and real-time data storage and management.
SR-03	Authentication Service	Firebase Authentication with OTP verification for secure user login and registration.
SR-04	Development Framework	React (Web), Flutter (Mobile) for cross-platform application development.
SR-05	API Services	RESTful APIs for communication between client and server modules.
SR-06	Mapping Service	Google Maps API integration for pharmacy location and navigation.
SR-07	Notification Service	Firebase Cloud Messaging (FCM) for real-time alerts and notifications.
SR-08	Security Tools	SSL/TLS protocols, encryption libraries, and role-based access control mechanisms.
SR-09	Backup & Recovery Software	Automated backup tools integrated with cloud storage for disaster recovery.
SR-10	Testing Tools	Postman, Selenium, and JUnit for API, functional, and unit testing.

Table 2.13: Software Requirements

2.6 Key Abstraction with CRC Analysis

CRC (Class–Responsibility–Collaborator) analysis is used to identify the main abstractions of the system, their responsibilities, and how they collaborate with one another. This ensures that the design is modular, scalable, and aligned with the system’s objectives.

2.6.1 Conceptual Modeling: Class Diagram

The conceptual class diagram represents the core entities of the **Medicine Price and Availability Finder Platform**, their attributes, and relationships.

Key Classes:

- **Patient:** Attributes include patient ID, name, contact, and medical history. Responsibilities include searching medicines, viewing prescriptions, and receiving notifications. Collaborates with Pharmacy and Prescription.
- **Pharmacy:** Attributes include pharmacy ID, name, location, and stock list. Responsibilities include updating medicine stock and prices, validating prescriptions. Collaborates with Medicine and Administrator.
- **Healthcare Professional:** Attributes include professional ID, name, specialization. Responsibilities include issuing prescriptions and updating patient records. Collaborates with Patient and Prescription.
- **Administrator:** Attributes include admin ID, name, role. Responsibilities include verifying pharmacies and healthcare professionals, overseeing system integrity. Collaborates with Pharmacy and Healthcare Professional.
- **Medicine:** Attributes include medicine ID, name, brand, price, stock quantity. Responsibilities include storing details and availability. Collaborates with Pharmacy and Patient.
- **Prescription:** Attributes include prescription ID, patient ID, medicine list, dosage, frequency. Responsibilities include storing verified prescription details. Collaborates with Patient and Healthcare Professional.
- **Notification:** Attributes include notification ID, type, timestamp. Responsibilities include alerting patients about medicine availability and price changes. Collaborates with Patient and System.

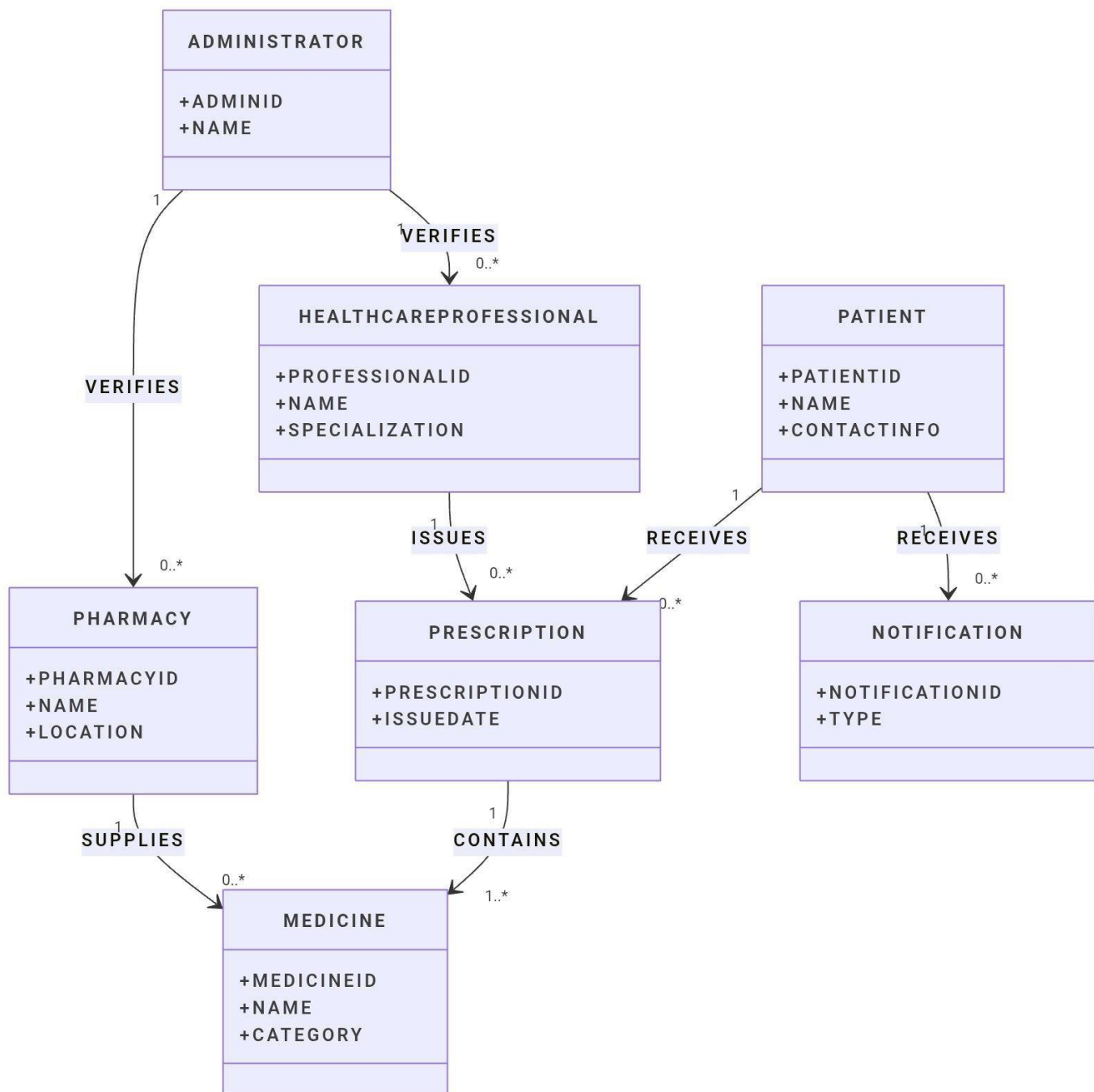


Figure 2.7: Conceptual Class Diagram – System Entities

2.6.2 Identifying Change Cases

Change cases represent potential modifications or extensions to the system in the future. They help anticipate scalability and adaptability.

Examples of Change Cases:

- **CC-1:** Addition of online medicine purchase and delivery functionality.
- **CC-2:** Integration with insurance systems for coverage and reimbursement.
- **CC-3:** Expansion to include laboratory test results and diagnostic reports.
- **CC-4:** Support for offline access in rural areas with poor connectivity.

- **CC-5:** Advanced analytics for medicine demand forecasting and supply chain optimization.
- **CC-6:** Integration with national health systems for centralized reporting.
- **CC-7:** AI-driven prescription error detection and patient guidance.

Change Case ID	Change Case	Description
CC-01	Add New Medicine	A pharmacy adds a new medicine to the system database with details such as name, category, price, and availability.
CC-02	Update Medicine Price	A pharmacy updates the price of an existing medicine to reflect market changes.
CC-03	Update Medicine Stock	A pharmacy modifies the stock quantity of a medicine to ensure accurate availability information.
CC-04	Remove Medicine	A pharmacy removes a medicine from the system when it is discontinued or permanently unavailable.
CC-05	Modify Prescription	A healthcare professional updates or corrects a patient's prescription details.
CC-06	Add Pharmacy	A new pharmacy registers in the system and awaits administrator verification.
CC-07	Verify Pharmacy	Administrator reviews and approves/rejects pharmacy registration details.
CC-08	Notify Patient	System sends notifications to patients about medicine availability, price changes, or prescription updates.
CC-09	Update User Profile	Patients or pharmacies update their account details (contact info, preferences, etc.).
CC-10	System Maintenance	Administrator performs updates, bug fixes, or enhancements to improve system performance and security.

Table 2.14: Change Cases

2.6.3 User Interface Prototyping

User interface prototypes demonstrate how the system will look and function for different actors.

Patient Dashboard:

- Search bar for medicines.
- List of available medicines with prices.
- Notifications panel for availability and price changes.
- Map integration showing nearby pharmacies.
- Tabs for Home, Prescriptions, Notifications, and Profile.

Pharmacy Dashboard:

- Stock update form for medicines.
- Price management interface.
- Prescription validation panel.
- Reports on medicine demand and sales.

Healthcare Professional Dashboard:

- Prescription issuance form.
- Patient record management interface.
- Tools for dosage and medication guidance.

Administrator Dashboard:

- Verification tools for pharmacies and healthcare professionals.
- System monitoring and reporting interface.
- User management and access control.

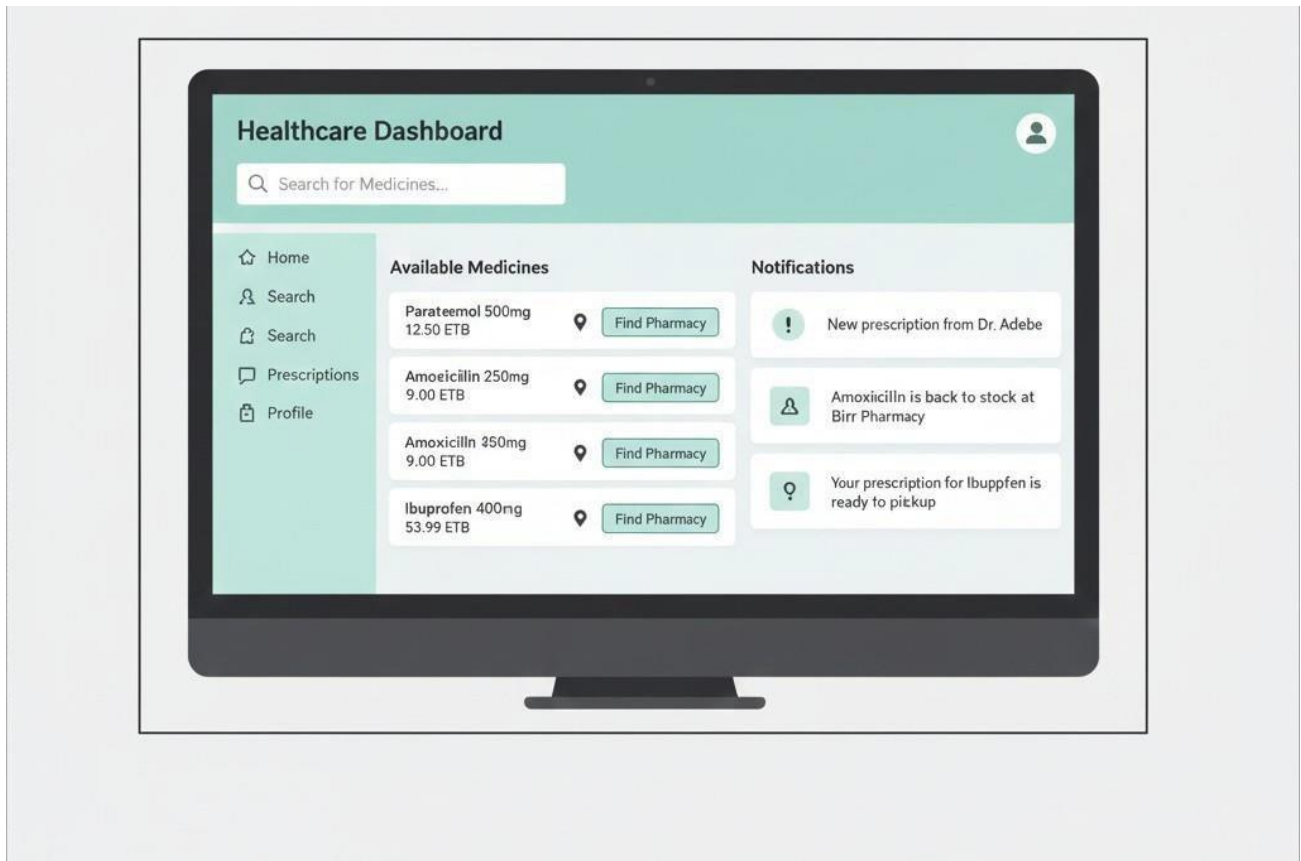


Figure 2.8: User Interface Prototype – Patient Dashboard

Chapter Three: System Design

This chapter presents a comprehensive and detailed design of the Medicine Price and Availability Finder Platform. The system design translates the requirements identified in Chapter Two into a structured technical blueprint that guides implementation. Emphasis is placed on clarity, modularity, scalability, and security, following Object-Oriented Design (OOD) principles and standard software engineering practices suitable for a Bachelor of Science in Information Technology project.

The purpose of this chapter is to explain **how the system is structured, how components interact, how data is stored and protected, and how users interact with the system** at a design level.

3.1 Architectural Design

The Medicine Price and Availability Finder Platform is designed using a **multi-tier layered architecture** combined with **cloud-based services**. This architectural style was selected because it separates system concerns into logical layers, making the system easier to understand, maintain, test, and extend in the future. In this architecture, each layer has a clearly defined responsibility and communicates with other layers through well-defined interfaces. This reduces dependency between components and improves system reliability.

- The layers found in this system are

Presentation Layer

This layer represents the user-facing part of the system. It includes mobile and web interfaces developed using Flutter. Patients, pharmacies, healthcare professionals, and administrators interact with the system through this layer. It is responsible for capturing user input, displaying data, and forwarding requests to the application layer.

Application Layer

The application layer contains the core business logic of the system. It processes user requests, enforces business rules, validates data, manages workflows such as medicine search and prescription issuance, and controls access based on user roles.

Data Layer

The data layer is responsible for storing and retrieving system data. Firebase Fire store is used as a NoSQL real-time database to store users, medicines, prescriptions, and notifications. Firebase Storage is used for storing documents and supporting files.

Integration Layer

This layer manages communication with external services. Google Maps API provides location and navigation services, while Firebase Cloud Messaging handles real-time notifications.

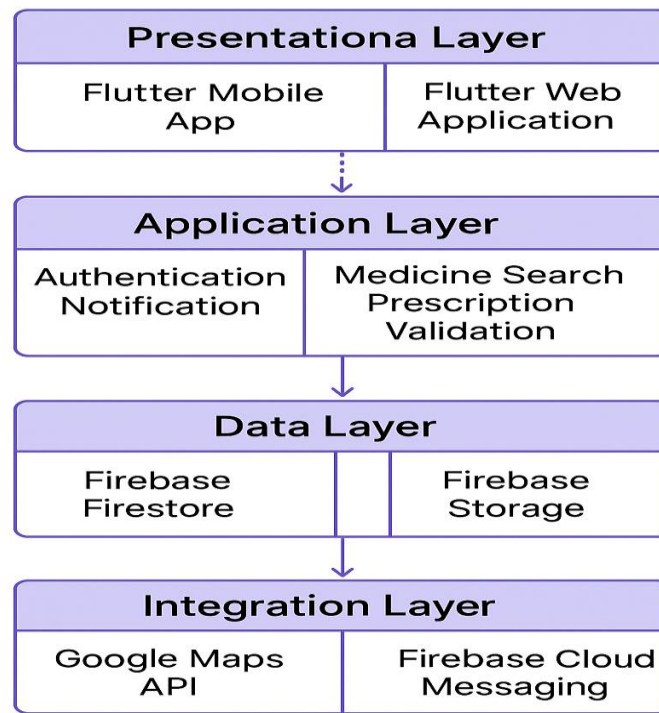


Figure 3.1: System Architecture Diagram

3.1.1 Component Modeling

Component modeling breaks the system into independent, reusable, and replaceable components. Each component performs a specific function and collaborates with others through defined interfaces.

User Interface Component

Handles screen rendering, form input, data display, and user interaction for all system roles.

Authentication Component

Manages OPT-based login, user session handling, and role identification using Firebase Authentication.

Medicine Management Component

Responsible for medicine search, price comparison, availability tracking, and filtering logic.

Prescription Management Component

Handles creation, validation, storage, and retrieval of digital prescriptions.

Notification Component

Monitors system events such as stock updates and prescription issuance and sends alerts to users.

Administration Component

Manages verification of pharmacies and healthcare professionals and oversees system integrity.

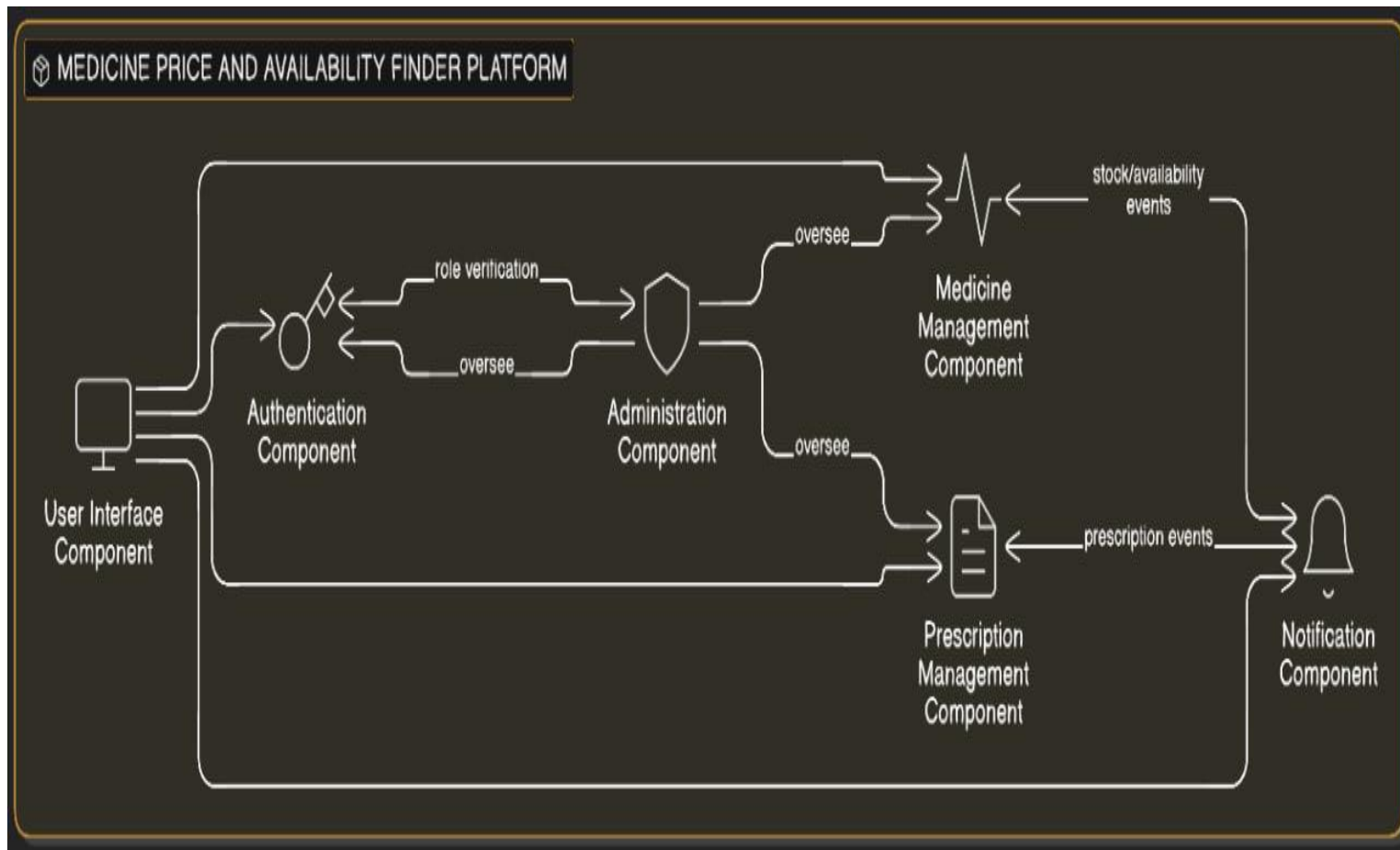


Figure 3.2: System component Diagram

3.1.2 Deployment Modeling

Deployment modeling describes how the system is physically deployed across hardware and cloud infrastructure. Users access the system using mobile devices or web browsers. All backend services are hosted on Firebase cloud infrastructure, eliminating the need for local servers.

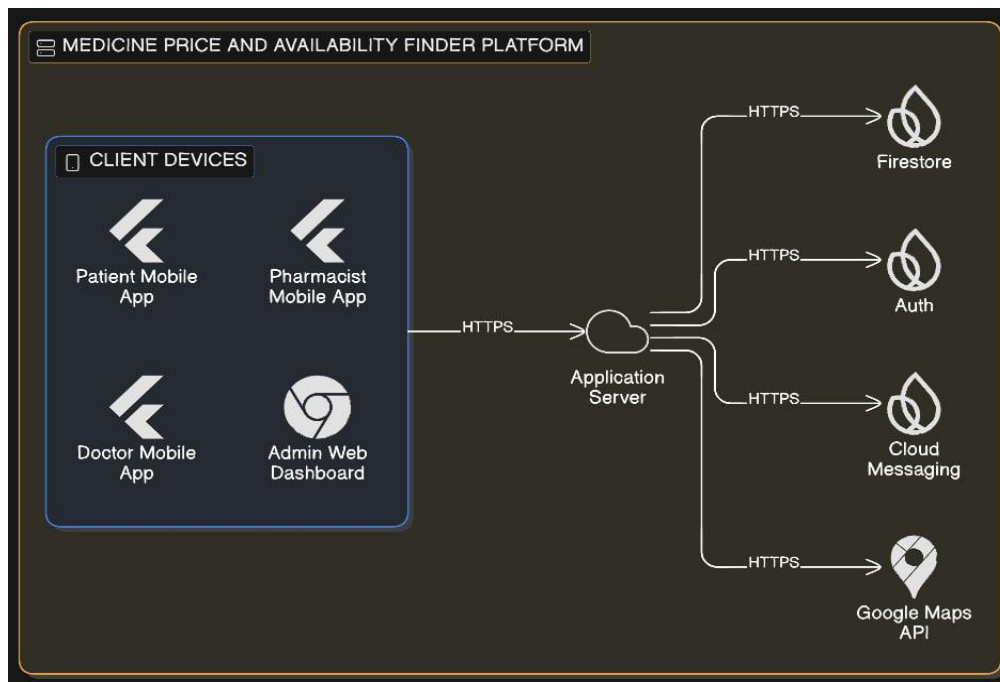


Figure 3.3: deployment diagram

This deployment model ensures

scalability, security, and minimal operational cost.

3.2 Detailed Design

Detailed design focuses on internal system structures, including class design and data persistence mechanisms.

3.2.1 Design Class Model

The design class model refines conceptual classes into implementable software classes. It defines attributes, responsibilities, and relationships.

User (Abstract Class)

Attributes: `userId`, `name`, `phoneNumber`, `role`

Responsibility: Common user properties and authentication handling

Patient

Attributes: `medicalHistory`

Responsibility: Search medicines, view prescriptions, receive notifications

Pharmacy

Attributes: `licenseNumber`, `location`

Responsibility: Update stock, validate prescriptions

HealthcareProfessional (inherits from User)

Inherited Attributes: `userId`, `name`, `phoneNumber`, `role`

Additional Attributes: `specialization`, `licenseId`

Responsibility: Issue and manage digital prescriptions linked to patients

Note on Inheritance: The `HealthcareProfessional` class inherits the `phoneNumber` attribute from the abstract `User` class. This is necessary because healthcare professionals must authenticate using OTP and receive system notifications, just like other users. Therefore, the phone number is not redefined but inherited to avoid redundancy.

Medicine

Attributes: `medicineId`, `name`, `brand`, `price`, `quantity`

Prescription

Attributes: `prescriptionId`, `dosage`, `frequency`, `duration`

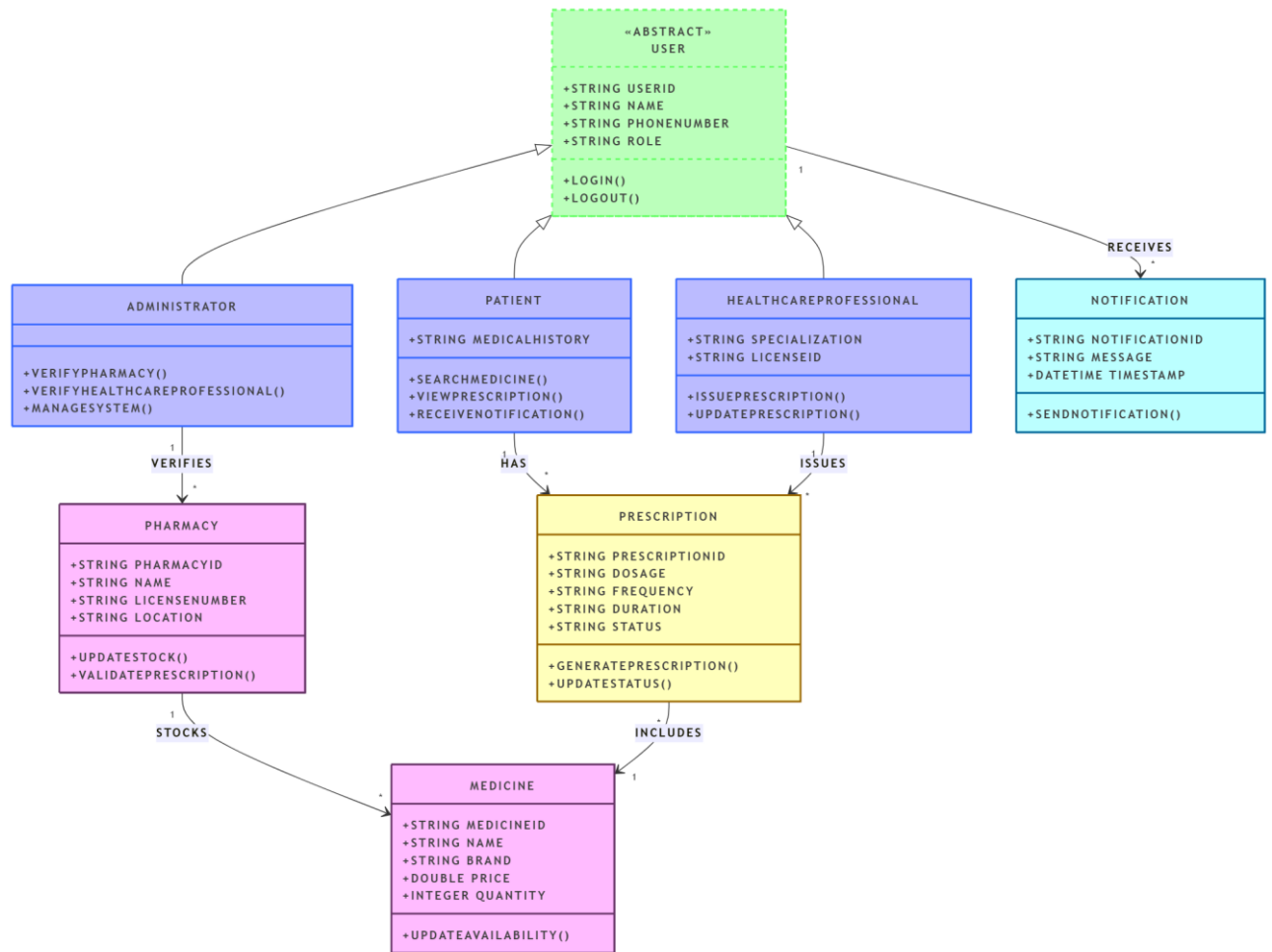


Figure 3.4: Design diagram

3.2.2 Persistent Model

The persistent model defines how data is stored in Firebase Firestore using collections and documents. Each entity is stored as a document with unique identifiers and timestamps.

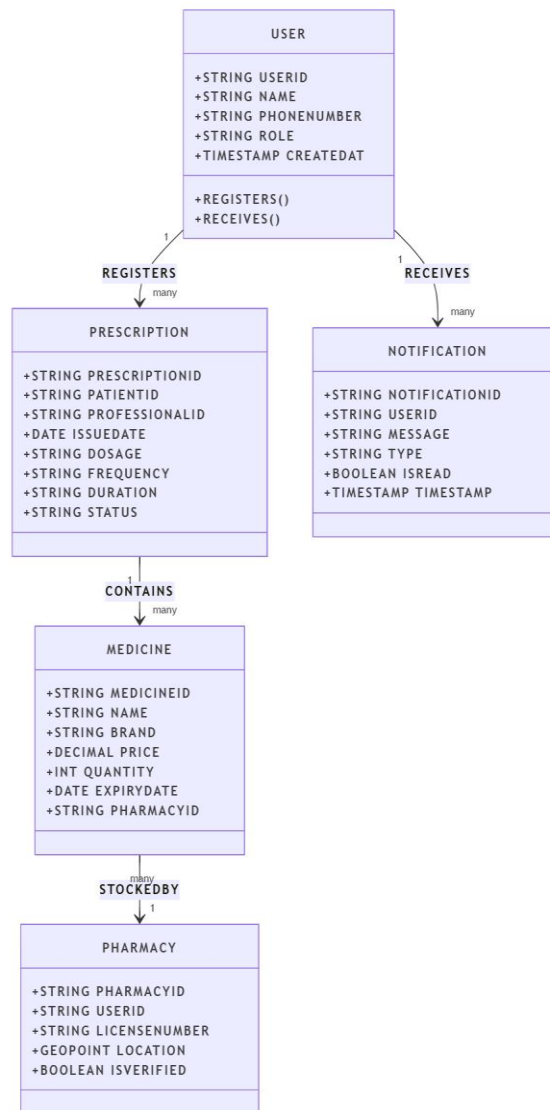


Figure 3.5: Persistent diagram

This model supports real-time updates and efficient querying.

3.3 User Interface Design

User interface design focuses on usability, accessibility, and consistency. The system follows a mobile-first approach while ensuring usability on web platforms.

UI Design Principles

- Simple navigation
- Minimal input steps
- Multilingual support
- Consistent layout across roles

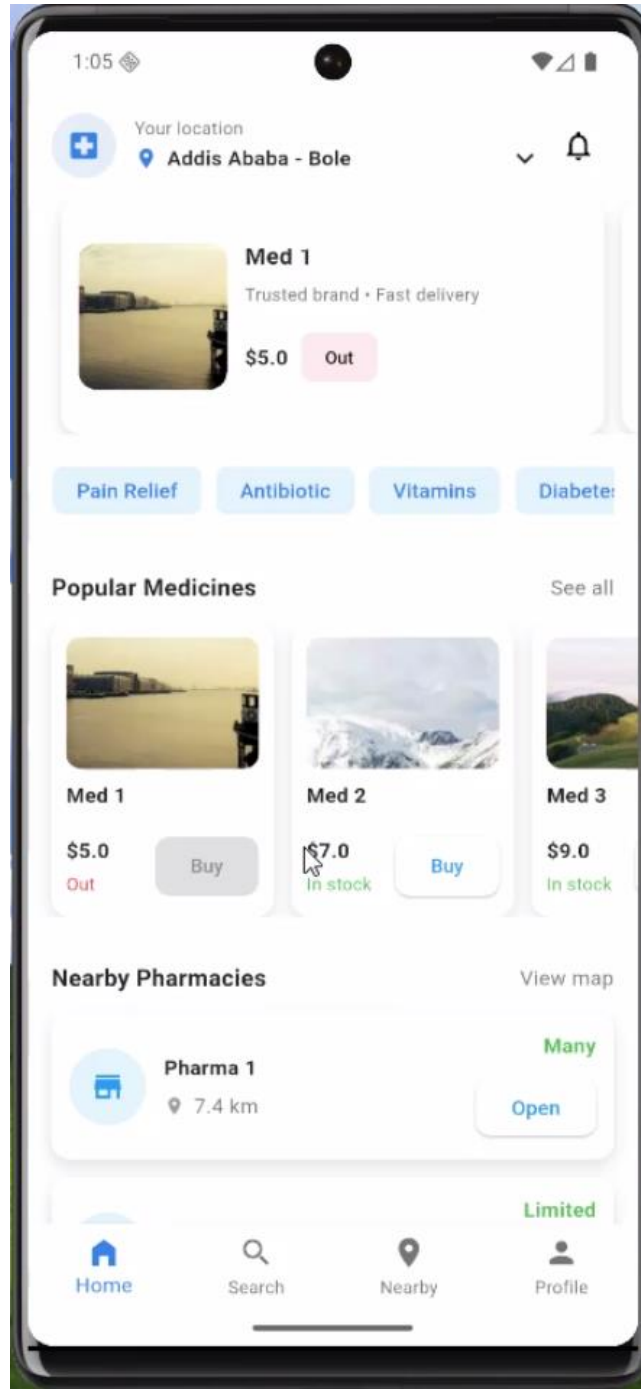


Figure 3.6: user-interface diagram

3.4 Access Control and Security

Access control and security are critical components of the Medicine Price and Availability Finder Platform because the system handles sensitive health-related data, including prescriptions, medicine availability, and user information. To ensure data confidentiality, integrity, and availability, the system implements **Role-Based Access Control (RBAC)** combined with secure authentication and authorization mechanisms.

RBAC ensures that each user can access only the system functions and data necessary for their role. This approach minimizes unauthorized access, prevents data misuse, and strengthens overall system security.

1. Access Control Levels (User Roles)

The system defines four primary access levels:

1. Patient
2. Pharmacy
3. Healthcare Professional
4. Administrator

Each role has clearly defined permissions and restrictions, as explained below.

2. Patient Access Level

Patients are end users who use the system to search for medicines, compare prices, and receive prescriptions. They are assigned the lowest privilege level because they should not modify medical or pharmacy data.

Permissions (What a Patient Can Do)

- Register and authenticate using OTP-based authentication
- Search for medicines by name or category
- View medicine prices and availability across multiple pharmacies
- View nearby pharmacies using map integration
- View their own prescriptions
- Receive notifications related to medicine availability and prescription issuance

Restrictions (What a Patient Cannot Do)

- Add, update, or delete medicine information
- Modify medicine prices or stock levels
- Issue or validate prescriptions
- Access other users' personal or medical data
- Perform administrative tasks

Security Justification

Restricting patients to read-only access ensures data integrity and prevents unauthorized modification of sensitive system data.

3. Pharmacy Access Level

Description

Pharmacies are authorized entities responsible for managing medicine inventory and validating prescriptions. Their access is restricted to data related to their own pharmacy only.

Permissions (What a Pharmacy Can Do)

- Authenticate using OTP after administrative verification
- Add new medicines to their inventory
- Update medicine prices and availability
- View prescriptions submitted by patients
- Validate or reject prescriptions
- Receive notifications related to prescriptions and stock updates

Restrictions (What a Pharmacy Cannot Do)

- Modify medicine data belonging to other pharmacies
- Issue prescriptions
- Access patient medical history
- Verify users or manage system-wide configurations

Security Justification

Limiting pharmacies to their own data prevents unfair competition, protects patient privacy, and ensures accountability.

4. Healthcare Professional Access Level

Description

Healthcare professionals are licensed individuals authorized to issue medical prescriptions. Their access is limited to prescription-related activities.

Permissions (What a Healthcare Professional Can Do)

- Authenticate using OTP after verification
- Issue digital prescriptions to patients
- View limited patient prescription history
- Select medicines when issuing prescriptions
- Receive notifications related to prescription status

Restrictions (What a Healthcare Professional Cannot Do)

- Modify medicine prices or stock levels
- Validate prescriptions
- Access pharmacy administrative data
- Manage user accounts or system settings

Security Justification

Restricting healthcare professionals to prescription-related actions ensures ethical compliance and prevents unauthorized system manipulation.

5. Administrator Access Level

Description

Administrators have the highest level of access and are responsible for system oversight, user verification, and policy enforcement.

Permissions (What an Administrator Can Do)

- Authenticate securely
- Verify and approve pharmacies and healthcare professionals
- Manage user accounts (activate, deactivate, or suspend)
- Monitor system activity and audit logs
- Enforce access control and security policies

Restrictions (What an Administrator Cannot Do)

- Issue medical prescriptions
- Update pharmacy stock or medicine prices unless required for maintenance

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Appendices

Appendix A: Sample Working Forms

- **Patient Registration Form:** Includes fields for patient ID, name, contact, and medical history.
- **Pharmacy Stock Update Form:** Fields for medicine name, brand, price, and quantity.
- **Prescription Form:** Issued by healthcare professionals with medicine details, dosage, frequency, and duration.

Appendix B: Questionnaire and Interview Questions

For Patients:

- How often do you face challenges finding prescribed medicines?
- Do you compare prices across pharmacies before purchasing?
- Would you use a mobile app to check medicine availability?

For Pharmacies:

- How do you currently manage stock and pricing records?
- What challenges do you face in updating medicine availability?
- Would you adopt a centralized digital platform for stock management?

For Healthcare Professionals:

- How do you issue and manage prescriptions?
- What common errors do you encounter in prescriptions?
- Would you prefer digital prescription management integrated with pharmacies?

Appendix C: Interview Guide for Pharmacists and Healthcare Professionals

- **Objective:** To understand workflows, challenges, and expectations regarding medicine availability and prescription management.
- **Key Topics:**
 - Current stock management practices.
 - Frequency and types of prescription errors.
 - Willingness to adopt digital systems.
 - Perceived benefits of real-time medicine availability platforms.

Appendix D: Screenshots of Prototype Interfaces

- **Patient Dashboard:** Search bar, medicine list, notifications, pharmacy map.
- **Pharmacy Dashboard:** Stock update form, price management, prescription validation.
- **Healthcare Professional Dashboard:** Prescription issuance and patient record management.
- **Administrator Dashboard:** Verification tools and system monitoring.

Appendix E: Sample Data Sets for Medicine Stock and Pricing

Medicine ID	Medicine Name	Brand	Price (ETB)	Stock Quantity	Pharmacy ID
M001	Paracetamol	MedTech	15	120	P001
M002	Amoxicillin	BioPharm	45	80	P002
M003	Ibuprofen	HealWell	25	200	P003
M004	Metformin	GlucoCare	60	50	P004
M005	Ciprofloxacin	PharmaLife	75	30	P005